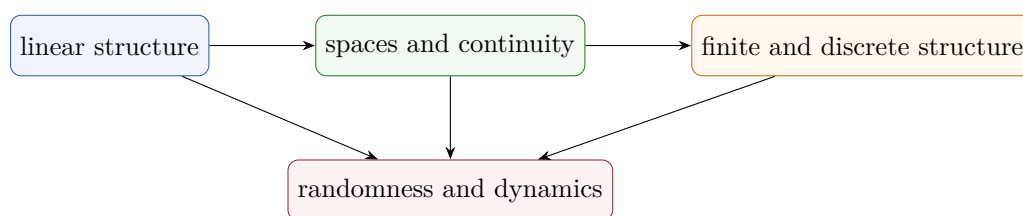


Mathematical Reboot II

Structures, Spaces, and Discrete Models

A self-contained refresher in linear algebra, multivariable calculus, topology, discrete mathematics, combinatorics, graph theory, number theory, probability, measure theory, and differential equations



Deep Intuition Edition

Designed for conceptual reconstruction before advanced study

August 2026

What this volume is for

This book is a second mathematical reboot: a concentrated reconstruction of subjects that often sit between an undergraduate mathematics degree and serious graduate work. It is not an encyclopedia and it is not a collection of exam tricks. Its purpose is to restore the web of ideas that makes advanced mathematics feel coherent rather than fragmented.

The governing question throughout is not merely “what is the definition?” but “what problem forced mathematicians to invent this definition, what mental picture makes it natural, which examples expose its boundaries, and what proof mechanism makes the main theorem work?” A concept is treated as learned only when one can move among its formal statement, its geometric or structural intuition, and concrete computations.

Deep intuition

A useful way to think about mathematical maturity is as the ability to change resolution. At low resolution you see a guiding picture: a linear map squashes some directions and preserves others; compactness prevents escape; a measure assigns consistent mass. At high resolution you can write the quantifiers, check hypotheses, and prove the theorem. Fluency means being able to move between those resolutions without losing the object.

The subjects and why they belong together

The chapters are organized around recurring structural themes.

- **Linear algebra** studies structure preserved by addition and scalar multiplication. It is the local language of calculus, differential equations, geometry, probability, optimization, and representation theory.
- **Multivariable calculus** explains why derivatives are linear maps and how local linear information controls nonlinear problems.
- **Topology** separates continuity from distance. It identifies the minimum structure needed to discuss nearness, convergence, compactness, and connectedness.
- **Discrete mathematics and combinatorics** develop the art of counting, recurrence, invariants, finite structure, and constructive proof.
- **Graph theory** turns relations into geometry and gives a laboratory for local-to-global reasoning.

- **Number theory** develops arithmetic structure through divisibility, congruences, and finite cyclic behavior.
- **Probability and measure theory** turn uncertainty and size into rigorous mathematics. Measure theory is also the natural completion of integration.
- **Differential equations and dynamical systems** ask how local rules generate global time evolution and show linearization in action.

How to use the book

For each major concept, use a five-pass reconstruction.

1. **Need.** State the mathematical problem the concept is designed to solve.
2. **Picture.** Give a mental model that is useful but not mistaken for the definition.
3. **Definition.** Write the exact formal statement.
4. **Boundary.** Produce an example, a nonexample, and a near-miss.
5. **Mechanism.** Reconstruct one theorem whose proof reveals why the definition has its exact form.

Common pitfall

Recognition is not retrieval. Reading a proof and thinking “yes, that makes sense” is much weaker than closing the book and rebuilding the argument. The exercises and recall drills are deliberately chosen so that passive familiarity is not enough.

Connection

Proof-scope convention. Central theorems used later in the book are proved in full. A small number of capstone results whose complete proof would require substantial additional machinery - most notably the uniqueness theory behind Jordan canonical form and the detailed verification that the outer-measure construction on $\mathbb{R}$ gives the usual Lebesgue length - are explicitly marked as proof architectures rather than being disguised as complete proofs.

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Diagnostic and dependency map

The point of a diagnostic is not to assign a grade. It is to locate the difference between *rust* and a structural gap. Rust means that the right object or theorem comes to mind but execution is slow. A gap means that the vocabulary itself no longer organizes the problem.

For each problem, mark C (confident), U (uncertain but a route is visible), or N (no workable route). Work without notes. Stop a problem after about fifteen minutes.

1.1 Diagnostic problems

Exercise 1.1 (Linear maps). Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be linear and suppose $\dim \ker T = 2$. Prove that $\dim \operatorname{im} T = 2$. Can T be surjective? Can it be injective? State the theorem you used in a basis-free form.

Exercise 1.2 (Eigenvalues and geometry). Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find its eigenvalues and eigenspaces. Is it diagonalizable over $\mathbb{R}$? Explain geometrically why repeated eigenvalue does not automatically mean “scalar multiplication by that eigenvalue.”

Exercise 1.3 (Orthogonal projection). Find the orthogonal projection of $(1, 2, 3)$ onto the plane $x + y + z = 0$. Derive the formula you use rather than quoting it.

Exercise 1.4 (Derivative as a linear map). For $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $F(x, y) = (x^2 - y, xy)$, compute $DF(1, 2)$. Explain what the derivative predicts about $F(1 + h, 2 + k)$ when (h, k) is small.

Exercise 1.5 (Compactness). Prove that a continuous function $f : [0, 1] \rightarrow \mathbb{R}$ is uniformly continuous. Identify exactly where compactness enters.

Exercise 1.6 (Topological continuity). Let X, Y be topological spaces. State continuity using inverse images of open sets. Show that this definition reduces to the familiar ε - δ definition for metric spaces.

Exercise 1.7 (Counting). How many surjections are there from a five-element set onto a three-element set? Give a derivation using inclusion-exclusion.

Exercise 1.8 (Graphs). Prove that every finite tree with at least two vertices has at least two leaves. Then prove that a graph is a tree if and only if it is connected and has $|V| - 1$ edges.

Exercise 1.9 (Congruences). Solve $17x \equiv 5 \pmod{43}$ and explain why a unique solution exists. Then state the criterion for $ax \equiv b \pmod{n}$ to be solvable.

Exercise 1.10 (Probability). If X and Y are independent square-integrable random variables, prove

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

What changes if independence is removed?

Exercise 1.11 (Measure). Give an example of a decreasing sequence of measurable sets E_n with $\mu(E_n) \rightarrow \infty$ for every n for which the naive identity

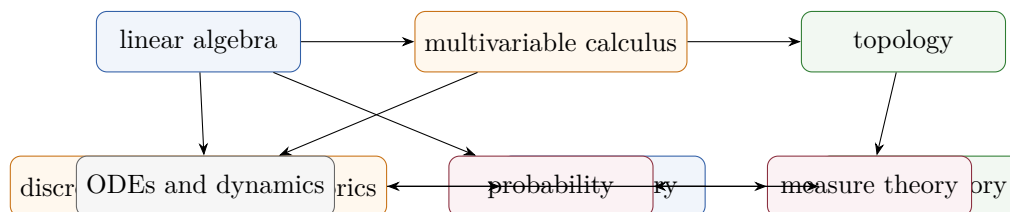
$$\mu\left(\bigcap_n E_n\right) = \lim_n \mu(E_n)$$

should make you suspicious. State the correct continuity-from-above theorem including its finiteness hypothesis.

Exercise 1.12 (Differential equations). Solve $x' = Ax$ when $A = \text{diag}(2, -1)$. Describe the phase portrait and explain how the signs of the eigenvalues determine local behavior.

1.2 How the subjects depend on one another

A useful map is not a prerequisite list but a network of reusable ideas.



Deep intuition

The deepest recurring pattern is *structure plus invariance*. Linear algebra asks what survives a change of basis. Topology asks what survives continuous deformation. Graph theory asks what global properties survive local rearrangement. Probability asks what remains stable under averaging. Differential equations ask what qualitative behavior survives perturbation. Learning the subjects together makes these repetitions visible.

1.3 Diagnostic interpretation

Pattern	Interpretation
Calculations work but definitions are vague	Rebuild the conceptual boxes and hypothesis audits before doing more exercises.
Definitions are known but proofs stall	Reconstruct proof architectures: kernel/image, compactness subsequences, induction invariants, conditioning, approximation by simple functions.
Linear algebra is weak	Repair it first. It is the local language of Chapters 3, 8, and 10.
Topology is weak	Spend extra time on quantifiers and examples. Many later theorems hide compactness or continuity assumptions.
Discrete topics feel like tricks	Focus on bijections, invariants, recursion, and extremal choices. These are reusable proof mechanisms, not isolated puzzles.
Probability feels intuitive but proofs are weak	Translate every verbal claim into events, sigma-algebras, or expectations before manipulating it.

Linear algebra I: spaces, maps, coordinates, and quotients

Linear algebra is often taught first as matrix manipulation. That is useful for computation but conceptually backwards. A matrix is a *coordinate description* of a linear transformation. The underlying objects are vector spaces and linear maps; bases are temporary measuring devices.

2.1 Why linear structure is so powerful

Motivation

Many mathematical problems become tractable when superposition is valid: if two inputs may be added and rescaled, and the output responds compatibly, then a complicated object can be decomposed into simple directions and rebuilt from them. Fourier analysis decomposes functions into frequencies, differential equations decompose motion into eigenmodes, statistics decomposes variation into orthogonal components, and differential geometry replaces a curved space locally by a tangent vector space.

Definition 2.1. A *vector space* over a field F is a set V with addition $V \times V \rightarrow V$ and scalar multiplication $F \times V \rightarrow V$ satisfying the usual associativity, commutativity, identity, inverse, and distributive axioms.

The axioms are not an arbitrary list. They guarantee that linear combinations

$$a_1v_1 + \cdots + a_nv_n$$

can be manipulated exactly as formal algebra suggests.

Deep intuition

A vector space is a world in which “direction,” “addition of effects,” and “change of scale” make sense. The elements need not be arrows. A polynomial, a matrix, a sequence, or a function can be a vector whenever those two operations behave linearly.

Example ladder: the same structure in different clothes

- $\mathbb{R}^n$ with ordinary coordinatewise operations.
- $P_n(\mathbb{R})$, polynomials of degree at most n . The monomials $1, x, \dots, x^n$ play the role of coordinate axes.
- $C([0, 1])$, continuous real-valued functions. This space is infinite-dimensional.
- $M_{m \times n}(F)$, matrices. The elementary matrices E_{ij} form a natural basis.
- The solution set of a homogeneous linear differential equation is a vector space; the solution set of an inhomogeneous equation generally is not.

2.2 Subspaces and span: which directions are actually available?

Definition 2.2. A subset $W \subseteq V$ is a *subspace* if it is itself a vector space under the inherited operations. Equivalently, W is nonempty and

$$u, v \in W, a, b \in F \implies au + bv \in W.$$

Proposition 2.3 (Subspace test). *A nonempty subset $W \subseteq V$ is a subspace if and only if it is closed under all combinations $au + bv$.*

Proof. Necessity is immediate. Conversely, closure under $au + bv$ gives $u + v \in W$ by taking $a = b = 1$, gives $au \in W$ by taking $v = u$ and choosing coefficients appropriately (or directly taking $b = 0$), gives $0 = 0u \in W$, and gives $-u = (-1)u \in W$. All remaining vector-space axioms are inherited from V . $\square$

Definition 2.4. For $S \subseteq V$, the *span* of S is the set of all finite linear combinations of elements of S :

$$\text{span}(S) = \left\{ \sum_{j=1}^n a_j s_j : n \geq 0, a_j \in F, s_j \in S \right\}.$$

Proposition 2.5. *$\text{span}(S)$ is the smallest subspace of V containing S .*

Proof. The zero vector is the empty linear combination, and sums and scalar multiples of finite linear combinations are finite linear combinations, so $\text{span}(S)$ is a subspace. It contains S . If W is any subspace containing S , closure of W under linear combinations forces every element of $\text{span}(S)$ to lie in W . Hence $\text{span}(S) \subseteq W$. $\square$

Deep intuition

Span is a *closure operation*. You begin with permitted directions and then close under the moves that linear structure allows. This viewpoint reappears everywhere: a subgroup generated by elements, an ideal generated by elements, the sigma-algebra generated by sets, and the topology generated by a basis are all versions of “take the smallest closed world containing the data.”

2.3 Linear independence and bases

Definition 2.6. Vectors $v_1, \dots, v_n$ are *linearly independent* if

$$a_1 v_1 + \dots + a_n v_n = 0$$

implies $a_1 = \dots = a_n = 0$. A *basis* is a linearly independent spanning set.

Motivation

Spanning asks whether we have enough building blocks. Independence asks whether any building block is redundant. A basis is exactly enough information: every vector can be built, and it can be built in only one way.

Proposition 2.7 (Uniqueness of coordinates). *If $(v_1, \dots, v_n)$ is a basis of V , then each $v \in V$ has a unique expression $v = \sum_i a_i v_i$.*

Proof. Existence follows from spanning. If $\sum_i a_i v_i = \sum_i b_i v_i$, then $\sum_i (a_i - b_i) v_i = 0$. Independence gives $a_i - b_i = 0$ for every i . $\square$

A basis is not a coordinate system until you choose it

In $P_2(\mathbb{R})$, both $(1, x, x^2)$ and $(1, x-1, (x-1)^2)$ are bases. The polynomial $p(x) = x^2$ has coordinates $(0, 0, 1)$ in the first basis but $(1, 2, 1)$ in the second because

$$x^2 = 1 + 2(x-1) + (x-1)^2.$$

The polynomial did not change; only its description did.

2.4 Dimension and the exchange mechanism

Lemma 2.8 (Steinitz exchange, finite form). *Suppose $v_1, \dots, v_m$ are linearly independent and $w_1, \dots, w_n$ span V . Then $m \leq n$, and after reordering, one can replace m of the w_j by the v_i and still obtain a spanning set.*

Proof. Because the w_j span, write $v_1 = \sum_j a_j w_j$. Some coefficient, say a_1 , is nonzero. Solve for w_1 in terms of $v_1, w_2, \dots, w_n$, so these new vectors span V . Now write v_2 in this spanning set. Its coefficient on at least one of $w_2, \dots, w_n$ must be nonzero; otherwise v_2 would be a multiple of v_1 , contradicting independence. Replace that w_j by v_2 . Continue. At the k th step there must remain an unreplaced w_j , so $k \leq n$. Thus $m \leq n$ and the stated exchange is possible. $\square$

Corollary 2.9. *Every two finite bases of the same vector space have the same number of elements.*

This common number is the *dimension* of V .

Proof architecture

The exchange lemma is the hidden engine behind most finite-dimensional dimension arguments. When a theorem compares the sizes of independent and spanning sets, mentally look for an exchange operation.

2.5 Linear maps: structure-preserving transformations

Definition 2.10. A map $T : V \rightarrow W$ is *linear* if

$$T(av + bw) = aT(v) + bT(w)$$

for all $v, w \in V$ and scalars a, b .

Deep intuition

A linear map does two things unusually well: it respects decomposition and it is completely determined by what it does to a basis. This is why linear models are computationally powerful. To know a nonlinear function on infinitely many points may require infinitely much information; to know a linear map on an n -dimensional space, n images suffice.

Proposition 2.11. *If $(v_1, \dots, v_n)$ is a basis of V , then a linear map $T : V \rightarrow W$ is uniquely determined by $T(v_1), \dots, T(v_n)$. Conversely, arbitrary choices $w_i \in W$ determine a unique linear map with $T(v_i) = w_i$.*

Proof. For $v = \sum a_i v_i$, linearity forces $T(v) = \sum a_i T(v_i)$, proving uniqueness. Conversely, define $T(\sum a_i v_i) = \sum a_i w_i$. Uniqueness of coordinates makes this well-defined, and the formula is visibly linear. $\square$

2.6 Kernel, image, and rank-nullity

Definition 2.12. For $T : V \rightarrow W$,

$$\ker T = \{v : T(v) = 0\}, \quad \text{im } T = \{T(v) : v \in V\}.$$

The *rank* is $\dim \text{im } T$ and the *nullity* is $\dim \ker T$.

Deep intuition

The kernel measures directions that the map completely collapses; the image measures directions that survive. Rank-nullity is a conservation law for dimension: every input direction is either visible at the output or lost in the kernel, with no overlap after an appropriate choice of basis.

Theorem 2.13 (Rank-nullity). *If V is finite-dimensional and $T : V \rightarrow W$ is linear, then*

$$\dim V = \dim \ker T + \dim \text{im } T.$$

Proof. Choose a basis $k_1, \dots, k_r$ of $\ker T$ and extend it to a basis

$$k_1, \dots, k_r, v_1, \dots, v_s$$

of V . We claim $T(v_1), \dots, T(v_s)$ is a basis of $\operatorname{im} T$. It spans: for $x = \sum a_i k_i + \sum b_j v_j$, we have $T(x) = \sum b_j T(v_j)$. It is independent: if $\sum b_j T(v_j) = 0$, then $\sum b_j v_j \in \ker T$, so it is a combination of the k_i . Since the combined list is independent, all $b_j = 0$. Thus $\dim \operatorname{im} T = s$ and $\dim V = r + s$. $\square$

Projection makes rank-nullity visible

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be projection onto the xy -plane: $T(x, y, z) = (x, y, 0)$. The kernel is the z -axis, dimension 1; the image is the xy -plane, dimension 2. The theorem says $3 = 1 + 2$. Geometrically, one direction disappears and two remain visible.

2.7 Matrices are coordinate representatives

Given bases $\mathcal{B} = (v_1, \dots, v_n)$ of V and $\mathcal{C} = (w_1, \dots, w_m)$ of W , the matrix $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ is formed by taking the coordinate vectors of $T(v_j)$ in $\mathcal{C}$ as columns.

Connection

Matrix multiplication is not an arbitrary rule. If $S : U \rightarrow V$ and $T : V \rightarrow W$, then coordinates give

$$[T \circ S] = [T][S].$$

The column-by-column multiplication algorithm is precisely the algebra required for composition of linear maps.

Common pitfall

Statements such as “ A is diagonalizable” are shorthand for statements about a linear operator after a basis has been chosen. Similar matrices represent the same operator in different bases. Matrix entries are not intrinsic; rank, determinant, trace, characteristic polynomial, and eigenvalues are intrinsic under change of basis.

2.8 Quotient spaces: declaring a subspace invisible

Definition 2.14. If $W \leq V$, define $v \sim v'$ when $v - v' \in W$. The *quotient space* V/W is the set of equivalence classes $v + W$, with

$$(v + W) + (u + W) = (v + u) + W, \quad a(v + W) = av + W.$$

Motivation

Suppose a measurement cannot detect changes in directions belonging to W . Then two vectors that differ by an element of W should count as the same observable state. The quotient space

is the mathematical world after those directions have deliberately been forgotten.

A three-dimensional world viewed from above

Let $V = \mathbb{R}^3$ and W be the vertical z -axis. In V/W , points on the same vertical line are identified. The quotient has dimension 2 and behaves like the xy -plane. What matters is not a preferred plane but the equivalence class of all points differing only vertically.

Proposition 2.15. *If V is finite-dimensional and $W \leq V$, then*

$$\dim(V/W) = \dim V - \dim W.$$

Proof. Choose a basis $w_1, \dots, w_r$ of W and extend it to $w_1, \dots, w_r, v_1, \dots, v_s$ of V . The cosets $v_1 + W, \dots, v_s + W$ span V/W because the w_i disappear in the quotient. If $\sum a_j(v_j + W) = W$, then $\sum a_j v_j \in W$, so $\sum a_j v_j$ is a combination of the w_i . Independence of the extended basis forces all $a_j = 0$. Hence these cosets form a basis and $s = \dim V - r$. $\square$

Theorem 2.16 (First isomorphism theorem for vector spaces). *For a linear map $T : V \rightarrow W$,*

$$V/\ker T \cong \operatorname{im} T.$$

Proof. Define $\tilde{T}(v + \ker T) = T(v)$. If $v + \ker T = v' + \ker T$, then $v - v' \in \ker T$, hence $T(v) = T(v')$, so the map is well-defined. It is linear and surjective onto $\operatorname{im} T$. If $\tilde{T}(v + \ker T) = 0$, then $v \in \ker T$, so $v + \ker T$ is the zero coset. Thus $\tilde{T}$ is injective. $\square$

Deep intuition

This theorem is rank-nullity with more structure. A linear map forgets exactly its kernel. Once vectors differing by a kernel direction are identified, no ambiguity remains, and the quotient is the image.

2.9 Dual spaces: measuring vectors linearly

Definition 2.17. The *dual space* V^* is the vector space of linear maps $\varphi : V \rightarrow F$. Its elements are called linear functionals.

Motivation

Vectors are often known indirectly through measurements. A sensor might report one coordinate, a weighted average, a moment, or a directional derivative. Linear functionals formalize linear measurements. The dual space is the collection of all such probes.

If $(v_1, \dots, v_n)$ is a basis, define $v^1, \dots, v^n \in V^*$ by $v^i(v_j) = \delta_{ij}$. This is the *dual basis*.

Proposition 2.18. *If V is finite-dimensional, then $\dim V^* = \dim V$, and the canonical map $J : V \rightarrow V^{**}$ given by*

$$J(v)(\varphi) = \varphi(v)$$

is an isomorphism.

Proof. The dual basis shows $\dim V^* = \dim V$. If $J(v) = 0$, then every functional vanishes on v . Writing $v = \sum a_i v_i$ and choosing v^j gives $a_j = 0$ for each j , hence $v = 0$. So J is injective. Since V and V^{**} have the same finite dimension, it is an isomorphism. $\square$

Common pitfall

The isomorphism $V \cong V^*$ is generally *not canonical*: it requires extra structure such as an inner product. By contrast, the map $V \rightarrow V^{**}$ above is canonical. This distinction between “isomorphic” and “naturally isomorphic” becomes increasingly important in advanced mathematics.

2.10 Recall drill

Recall drill

Without looking back, explain: (1) span as a closure operation; (2) why a basis gives unique coordinates; (3) the proof mechanism of rank-nullity; (4) what a quotient space forgets; (5) why the first isomorphism theorem is really a statement about information loss; (6) what a dual vector measures.

2.11 Exercises

Exercise 2.1. Let $T : P_3(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be differentiation. Find bases for $\ker T$ and $\operatorname{im} T$, verify rank-nullity, and identify $P_3/\ker T$ concretely with P_2 .

Exercise 2.2. Prove that the solution set of $Ax = b$ is either empty or an affine translate $x_0 + \ker A$. Explain why it is usually not a subspace when $b \neq 0$.

Exercise 2.3. Let $U, W \leq V$ with V finite-dimensional. Prove

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Hint: use the map $U \oplus W \rightarrow U + W$, $(u, w) \mapsto u + w$.

Exercise 2.4. If $T : V \rightarrow W$ is injective and V is finite-dimensional, prove that any linearly independent set in V has linearly independent image. State and prove the dual statement for spanning sets under surjective maps.

Exercise 2.5. Let $W = \{(x, y, z) : x + y + z = 0\} \subset \mathbb{R}^3$. Find a basis of the annihilator $W^\circ = \{\varphi \in (\mathbb{R}^3)^* : \varphi|_W = 0\}$ and explain geometrically what it measures.

Linear algebra II: determinants, eigen-structure, orthogonality, and canonical form

The first chapter treated linear maps without privileging coordinates. We now ask what features of a linear operator remain unchanged under a change of basis and how those invariants reveal geometry and dynamics.

3.1 Determinants: oriented volume scaling

Motivation

A linear map can stretch different directions by different factors, shear, reflect, or collapse dimension. We want one scalar that records the total signed volume change. That scalar is the determinant.

For an n -dimensional vector space, the determinant can be characterized as the unique alternating multilinear function of the columns satisfying $\det I = 1$.

Deep intuition

Multilinearity says that volume responds linearly when one edge of a parallelepiped is varied while the others are fixed. Alternation says volume vanishes when two edge directions coincide and flips sign when two edges are exchanged. These are exactly the geometric rules for oriented volume.

Theorem 3.1. *For square matrices A, B over a field,*

$$\det(AB) = \det(A) \det(B).$$

Consequently, A is invertible if and only if $\det A \neq 0$.

Proof. Fix B and regard the columns of AB as A applied to the columns of B . The function $D(B) = \det(AB)$ is alternating and multilinear in the columns of B . Hence, by uniqueness of the

determinant up to its value on the identity,

$$D(B) = D(I) \det B = \det A \det B.$$

If A is invertible, $1 = \det I = \det A \det A^{-1}$, so $\det A \neq 0$. Conversely, if $\det A \neq 0$, the columns cannot be dependent, because an alternating multilinear determinant vanishes on dependent columns. Hence the columns form a basis and A is invertible. $\square$

Shear versus scaling

The shear

$$S = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$$

can move a square to a slanted parallelogram while preserving area, so $\det S = 1$. The diagonal map $D = \text{diag}(a, b)$ scales coordinate directions independently, so $\det D = ab$. A reflection has negative determinant because it reverses orientation.

3.2 Trace and determinant as similarity invariants

If $B = P^{-1}AP$, then

$$\det B = \det A, \quad \text{tr } B = \text{tr } A.$$

The determinant statement follows from multiplicativity. For trace, the identity $\text{tr}(XY) = \text{tr}(YX)$ gives $\text{tr}(P^{-1}AP) = \text{tr}(APP^{-1}) = \text{tr } A$.

Connection

Similarity means “same linear operator, different basis.” Any quantity preserved by similarity is a candidate for intrinsic information about the operator. The characteristic polynomial packages many such invariants at once.

3.3 Eigenvectors: directions that do not turn

Definition 3.2. For $T : V \rightarrow V$, a nonzero vector v is an *eigenvector* with eigenvalue λ if

$$T(v) = \lambda v.$$

The eigenspace is $E_\lambda = \ker(T - \lambda I)$.

Deep intuition

An arbitrary vector can change both direction and length under T . An eigenvector is a direction on which the complicated transformation becomes one-dimensional: only a scalar remains. Eigenvectors are therefore the natural coordinates for repeated application of T .

Proposition 3.3. *Eigenvectors associated with distinct eigenvalues are linearly independent.*

Proof. Suppose $v_1, \dots, v_k$ correspond to distinct λ_i and choose a shortest nontrivial dependence $\sum_i a_i v_i = 0$. Apply T and subtract λ_k times the original relation:

$$\sum_{i=1}^{k-1} a_i (\lambda_i - \lambda_k) v_i = 0.$$

By minimality of the original relation, every coefficient is zero. Since $\lambda_i \neq \lambda_k$, $a_i = 0$ for $i < k$, and then $a_k = 0$, contradiction. $\square$

Definition 3.4. The characteristic polynomial of A is $\chi_A(t) = \det(tI - A)$. Its roots are exactly the eigenvalues.

3.4 Diagonalization: decoupling the transformation

Theorem 3.5. An operator T on an n -dimensional space is diagonalizable if and only if V has a basis consisting of eigenvectors of T .

Proof. If an eigenbasis (v_i) exists, the matrix of T in that basis has λ_i in the i th diagonal position and zeros elsewhere. Conversely, if some basis gives a diagonal matrix, the basis vectors are visibly eigenvectors. $\square$

Motivation

A diagonal matrix turns one coupled n -dimensional problem into n independent one-dimensional problems. Powers become $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$, exponentials become $e^{tD} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$, and quadratic forms become sums of independent squares after orthogonal diagonalization.

Why one eigenvalue may be insufficient

For

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix},$$

$\chi_A(t) = (t - 2)^2$, but

$$A - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has one-dimensional kernel. There is only one independent eigenvector, so A is not diagonalizable. The missing direction is not scaled independently; it is sheared into the eigenvector direction.

Common pitfall

Algebraic multiplicity counts how many times an eigenvalue occurs as a root of χ_A . Geometric multiplicity is $\dim E_\lambda$. Always

$$1 \leq \dim E_\lambda \leq \text{algebraic multiplicity of } \lambda.$$

Diagonalizability requires the total geometric multiplicity to reach n .

3.5 Inner products: geometry added to linear algebra

Definition 3.6. An inner product on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ that is bilinear, symmetric, and positive definite. It defines

$$\|v\| = \sqrt{\langle v, v \rangle}, \quad \cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

Theorem 3.7 (Cauchy-Schwarz). *For all u, v in an inner-product space,*

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds exactly when u, v are linearly dependent.

Proof. If $v = 0$ the claim is immediate. Otherwise, for every real t ,

$$0 \leq \|u - tv\|^2 = \|u\|^2 - 2t \langle u, v \rangle + t^2 \|v\|^2.$$

The quadratic in t has nonpositive discriminant, so

$$4 \langle u, v \rangle^2 - 4 \|u\|^2 \|v\|^2 \leq 0.$$

Equality means the quadratic has a zero, hence $u - tv = 0$ for some t . □

Deep intuition

Cauchy-Schwarz is the statement that a component of a vector in a chosen direction cannot be longer than the vector itself. The algebraic proof by minimizing $\|u - tv\|$ is literally the geometry of finding the closest point to u on the line through v .

3.6 Orthogonal projection and least squares

Let $W \leq V$ be finite-dimensional with orthonormal basis $e_1, \dots, e_k$. Define

$$P_W v = \sum_{i=1}^k \langle v, e_i \rangle e_i.$$

Then $v - P_W v$ is orthogonal to every e_i , hence to all of W .

Theorem 3.8 (Best approximation). *$P_W v$ is the unique vector in W minimizing $\|v - w\|$.*

Proof. For $w \in W$,

$$v - w = (v - P_W v) + (P_W v - w),$$

and the two summands are orthogonal. Pythagoras gives

$$\|v - w\|^2 = \|v - P_W v\|^2 + \|P_W v - w\|^2 \geq \|v - P_W v\|^2,$$

with equality only when $w = P_W v$. □

Least squares is projection

For an inconsistent system $Ax = b$, the least-squares problem minimizes $\|Ax - b\|$. Since Ax ranges over the column space $\mathcal{C}(A)$, we are projecting b onto $\mathcal{C}(A)$. Orthogonality of the residual gives

$$A^T(Ax - b) = 0,$$

so the normal equations are $A^T Ax = A^T b$. They are not a mysterious algebraic trick; they are the equation “residual is perpendicular to every column direction.”

3.7 Gram-Schmidt and QR factorization

Given independent vectors $v_1, \dots, v_n$, define recursively

$$u_1 = v_1, \quad u_k = v_k - \sum_{j < k} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j,$$

and normalize $e_k = u_k / \|u_k\|$.

Proposition 3.9. *The resulting $e_1, \dots, e_n$ are orthonormal and satisfy*

$$\text{span}(e_1, \dots, e_k) = \text{span}(v_1, \dots, v_k)$$

for each k .

Proof. By construction, u_k is obtained from v_k by subtracting its projections onto earlier u_j , hence is orthogonal to them. Independence of the v_i ensures $u_k \neq 0$. The span equality follows in both directions from the recursive formula: u_k lies in the span of the first k v ’s, while v_k equals u_k plus a combination of earlier u_j ’s. $\square$

3.8 Adjoints and self-adjoint operators

Definition 3.10. For finite-dimensional inner-product spaces, the adjoint T^* is defined by

$$\langle Tv, w \rangle = \langle v, T^*w \rangle$$

for all v, w . In an orthonormal basis over $\mathbb{R}$, $[T^*] = [T]^T$; over $\mathbb{C}$, it is the conjugate transpose.

An operator is *self-adjoint* if $T = T^*$.

Theorem 3.11 (Spectral theorem, finite-dimensional real form). *Every self-adjoint operator on a finite-dimensional real inner-product space admits an orthonormal basis of eigenvectors. Equivalently, every real symmetric matrix is orthogonally diagonalizable.*

Proof. We argue by induction on $n = \dim V$. Choose an orthonormal basis and let A be the real symmetric matrix of T . Its characteristic polynomial has a complex root λ and a nonzero complex eigenvector z . Using the Hermitian inner product on $\mathbb{C}^n$,

$$\lambda z^* z = z^* A z.$$

Because $A = A^T = A^*$, the scalar z^*Az is real: its complex conjugate is $z^*A^*z = z^*Az$. Since $z^*z > 0$, λ is real. Writing $z = x + iy$ with $x, y \in \mathbb{R}^n$, the equation $Az = \lambda z$ gives $Ax = \lambda x$ and $Ay = \lambda y$; at least one of x, y is nonzero. Hence T has a real unit eigenvector e_1 .

Its orthogonal complement $W = e_1^\perp$ is T -invariant, because for $w \in W$,

$$\langle Tw, e_1 \rangle = \langle w, Te_1 \rangle = \lambda \langle w, e_1 \rangle = 0.$$

The restriction $T|_W$ is self-adjoint. By induction, W has an orthonormal eigenbasis, and adjoining e_1 gives an orthonormal eigenbasis of V . $\square$

Deep intuition

Symmetry of a matrix is not cosmetic. It says the transformation has no hidden rotational mismatch between input and output directions. The spectral theorem is the reward: there is an orthonormal coordinate system in which the transformation is pure independent scaling.

3.9 Positive definite forms and quadratic geometry

A symmetric matrix A defines $q(x) = x^T Ax$. It is *positive definite* if $q(x) > 0$ for every $x \neq 0$.

By the spectral theorem, $A = QDQ^T$ with Q orthogonal, so setting $y = Q^T x$ gives

$$q(x) = \sum_i \lambda_i y_i^2.$$

Thus A is positive definite exactly when every eigenvalue is positive.

Ellipses are diagonal quadratic forms in disguise

The level set $x^T Ax = 1$ for positive-definite A is an ellipsoid. The eigenvectors give its principal axes and the semi-axis lengths are $1/\sqrt{\lambda_i}$. Linear algebra turns a tilted quadratic equation into independent one-dimensional scales.

3.10 Optional capstone: Jordan form when diagonalization fails

Over $\mathbb{C}$, every operator can be represented in Jordan form. A Jordan block

$$J_\lambda = \lambda I + N, \quad N = \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & & \ddots & 1 \\ 0 & \cdots & 0 & 0 \end{pmatrix}$$

contains a diagonal scaling part and a nilpotent shear part.

A full proof of Jordan form is longer than the other proofs in this chapter, so we isolate the mechanism.

Connection

Jordan canonical form. Let T act on a finite-dimensional complex vector space. Then some basis gives a block diagonal matrix whose blocks are Jordan blocks. The multiset of block sizes for each eigenvalue is uniquely determined up to order.

Proof architecture

Factor the minimal polynomial into $(t - \lambda_i)^{m_i}$. The primary decomposition theorem splits

$$V = \bigoplus_i \ker(T - \lambda_i I)^{m_i}.$$

On each summand, subtract $\lambda_i I$ and study a nilpotent operator N . A nilpotent map organizes vectors into chains

$$v, Nv, N^2v, \dots, N^{r-1}v,$$

ending at zero. Choosing a basis made of such chains produces Jordan blocks. The block sizes are recovered from the dimensions $\dim \ker N^k$, so they are intrinsic.

Connection

Jordan form explains polynomial factors such as $t^k e^{\lambda t}$ in linear differential equations. The exponential of a Jordan block is

$$e^{t(\lambda I + N)} = e^{\lambda t} \left(I + tN + \frac{t^2 N^2}{2!} + \dots \right),$$

and the series terminates because N is nilpotent.

3.11 Recall drill

Recall drill

Explain: determinant as volume scaling; why eigenvectors matter for iteration; the difference between algebraic and geometric multiplicity; why least squares is projection; the geometric meaning of self-adjointness; what the nilpotent part of a Jordan block represents.

3.12 Exercises

Exercise 3.1. Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ directly from entries. Deduce that trace is invariant under similarity.

Exercise 3.2. For $A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$, find an orthonormal eigenbasis, determine whether A is positive definite, and describe the level set $x^T A x = 1$.

Exercise 3.3. Let $W = \text{span}\{(1, 1, 0), (0, 1, 1)\} \subset \mathbb{R}^3$. Compute the orthogonal projection matrix onto W by first orthonormalizing a basis.

Exercise 3.4. Prove that if T is self-adjoint, eigenspaces for distinct eigenvalues are orthogonal.

Exercise 3.5. Compute A^n for $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ without multiplying repeatedly. Interpret the polynomial factor in n using the nilpotent part.

Multivariable calculus: derivatives as local linear models

Single-variable calculus can hide the true nature of differentiation because the derivative is just a number. In several variables the central fact becomes unavoidable: a derivative is a linear map that gives the best first-order approximation to a nonlinear map.

4.1 From slope to linear approximation

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We say F is differentiable at a if there is a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$F(a + h) = F(a) + Ah + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0 \quad (h \rightarrow 0).$$

The map A , necessarily unique, is denoted $DF(a)$.

Motivation

A derivative should be the object that remains after we zoom in far enough that nonlinear curvature becomes negligible. In one variable, the only linear maps $\mathbb{R} \rightarrow \mathbb{R}$ are multiplication by a number. In several variables, the correct replacement for “slope” is therefore a genuine linear transformation.

Deep intuition

Differentiability says the error after subtracting the linear model is *smaller than first order*. The condition $\|r(h)\| / \|h\| \rightarrow 0$ is stronger than merely saying $r(h) \rightarrow 0$. It says that under magnification by the factor $1/\|h\|$, the error disappears while the linear part remains visible.

Proposition 4.1 (Uniqueness of the derivative). *If A and B both satisfy the differentiability condition at a , then $A = B$.*

Proof. For every h ,

$$(A - B)h = o(\|h\|).$$

Fix v and put $h = tv$. Then

$$\|(A - B)v\| = \frac{1}{|t|} \|(A - B)(tv)\| = \frac{o(|t| \|v\|)}{|t|} \rightarrow 0.$$

Hence $(A - B)v = 0$ for all v , so $A = B$. □

4.2 The Jacobian is a matrix representation of the derivative

If all first partial derivatives exist near a and are continuous at a , then F is differentiable at a , and in standard coordinates

$$[DF(a)] = J_F(a) = \left(\frac{\partial F_i}{\partial x_j}(a) \right)_{i,j}.$$

A nonlinear map seen through its tangent map

Let $F(x, y) = (x^2 - y, xy)$. Then

$$J_F(x, y) = \begin{pmatrix} 2x & -1 \\ y & x \end{pmatrix}, \quad J_F(1, 2) = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}.$$

Thus for small (h, k) ,

$$F(1 + h, 2 + k) \approx F(1, 2) + \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix}.$$

The matrix is not just a table of partial derivatives: it is the best linear machine predicting how small input displacements move at the output.

Common pitfall

Existence of all partial derivatives at a point does not imply differentiability there. Partial derivatives only inspect coordinate directions. Differentiability demands one linear approximation that works uniformly over *all* directions of approach.

Partial derivatives without differentiability

Define

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Along each coordinate axis, $f = 0$, so both partial derivatives at the origin exist and are zero. But along $y = x^2$,

$$f(x, x^2) = \frac{x^4}{2x^4} = \frac{1}{2},$$

so f is not even continuous at the origin, hence cannot be differentiable.

4.3 Chain rule: local linear models compose

Theorem 4.2 (Chain rule). *If $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at a and $G : \mathbb{R}^m \rightarrow \mathbb{R}^k$ is differentiable at $F(a)$, then $G \circ F$ is differentiable at a and*

$$D(G \circ F)(a) = DG(F(a)) \circ DF(a).$$

Proof. Write

$$F(a + h) = F(a) + Ah + r(h), \quad \|r(h)\| = o(\|h\|),$$

and

$$G(F(a) + u) = G(F(a)) + Bu + s(u), \quad \|s(u)\| = o(\|u\|),$$

where $A = DF(a)$ and $B = DG(F(a))$. Then

$$G(F(a + h)) - G(F(a)) = B(Ah + r(h)) + s(Ah + r(h)).$$

Subtract BAh . The error is $Br(h) + s(Ah + r(h))$. The first term is $o(\|h\|)$ because linear maps are bounded in finite dimension. Also $\|Ah + r(h)\| = O(\|h\|)$, so the second term is $o(\|h\|)$. Therefore BA is the derivative. $\square$

Deep intuition

The chain rule is inevitable once derivatives are understood as best linear approximations. First zoom into F , then zoom into G . At first order, composing the nonlinear maps is indistinguishable from composing their tangent linear maps. Matrix multiplication appears because it represents composition.

4.4 Directional derivatives, gradient, and the geometry of change

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at a , the derivative is a linear functional. With the Euclidean inner product, every linear functional has the form

$$Df(a)h = \langle \nabla f(a), h \rangle,$$

where $\nabla f(a)$ is the gradient.

For a unit vector u , the directional derivative is

$$D_u f(a) = \langle \nabla f(a), u \rangle.$$

Cauchy-Schwarz yields

$$D_u f(a) \leq \|\nabla f(a)\|,$$

with equality when u points in the gradient direction.

Deep intuition

The gradient is not “a vector of partial derivatives” by accident. The derivative $Df(a)$ is naturally a covector: it eats a displacement and returns first-order change. The inner product

converts that covector into the vector $\nabla f(a)$. This is why gradients depend on the metric even though differentials do not.

4.5 Second derivatives and the Hessian

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable, its second derivative is a bilinear form. In coordinates it is represented by the Hessian

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right).$$

When second partials are continuous, Clairaut's theorem gives symmetry.

Taylor's theorem at second order reads

$$f(a+h) = f(a) + Df(a)h + \frac{1}{2}h^T H_f(a)h + o(\|h\|^2).$$

At a critical point where $\nabla f(a) = 0$, the quadratic form $h^T H_f(a)h$ gives the first nontrivial local shape.

Theorem 4.3 (Second derivative test). *Suppose f is C^2 near a critical point a .*

1. *If $H_f(a)$ is positive definite, then a is a strict local minimum.*
2. *If $H_f(a)$ is negative definite, then a is a strict local maximum.*
3. *If $H_f(a)$ is indefinite, then a is a saddle point.*

Proof. For the positive definite case, let $\lambda_{\min} > 0$ be the smallest eigenvalue of the symmetric Hessian. Then $h^T H_f(a)h \geq \lambda_{\min} \|h\|^2$. Taylor's formula gives

$$f(a+h) - f(a) \geq \frac{1}{2}\lambda_{\min} \|h\|^2 + o(\|h\|^2),$$

which is positive for sufficiently small nonzero h . The negative definite case follows by applying this to $-f$. If the Hessian is indefinite, there are directions u, v with $u^T H u > 0$ and $v^T H v < 0$. Along the lines $a + tu$ and $a + tv$, Taylor's formula gives values respectively above and below $f(a)$ for small $t \neq 0$. $\square$

Common pitfall

A semidefinite Hessian gives no conclusion. For example, $f(x) = x^4$ has Hessian 0 at the origin but a strict minimum, whereas $g(x) = x^3$ also has second derivative 0 there but no extremum.

4.6 Constrained optimization and Lagrange multipliers

Suppose we want to optimize f subject to $g(x) = c$. If $\nabla g(a) \neq 0$, then near a the constraint surface has tangent space

$$T_a M = \{h : Dg(a)h = 0\} = \nabla g(a)^\perp.$$

At a constrained extremum, $Df(a)$ must vanish on all tangent directions, so $\nabla f(a)$ is orthogonal to $T_a M$. Hence it lies in the one-dimensional normal direction spanned by $\nabla g(a)$.

Theorem 4.4 (Lagrange multiplier condition). *If a is a local extremum of f restricted to the smooth level set $g = c$ and $\nabla g(a) \neq 0$, then there is λ such that*

$$\nabla f(a) = \lambda \nabla g(a).$$

Proof. Because $Dg(a) \neq 0$, after relabeling coordinates we may assume $\partial g / \partial x_n(a) \neq 0$. The implicit function theorem then parameterizes the level set near a by the first $n - 1$ coordinates. Equivalently, every tangent vector h to the level set satisfies $Dg(a)h = 0$, and every vector in $\ker Dg(a)$ occurs as the velocity of a smooth curve in the level set through a .

If γ is such a curve, then $f \circ \gamma$ has a local extremum at 0, so

$$0 = (f \circ \gamma)'(0) = Df(a)\gamma'(0).$$

Hence $Df(a)$ vanishes on $\ker Dg(a)$. Since $Dg(a)$ is a nonzero linear functional, choose v with $Dg(a)v = 1$. Any h decomposes as

$$h = (h - Dg(a)hv) + Dg(a)hv,$$

where the first term lies in $\ker Dg(a)$. Therefore

$$Df(a)h = Df(a)v Dg(a)h.$$

Thus $Df(a) = \lambda Dg(a)$ with $\lambda = Df(a)v$. Under the Euclidean inner product, this is exactly $\nabla f(a) = \lambda \nabla g(a)$. $\square$

Closest point to a linearly constrained surface

Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 1$. Then

$$2(x, y, z) = \lambda(1, 1, 1),$$

so $x = y = z$. The constraint gives $x = y = z = 1/3$. Geometrically, the shortest radius to a plane is perpendicular to the plane.

4.7 Inverse function theorem: when a nonlinear map has local coordinates

Motivation

If the derivative $DF(a)$ is invertible, the local linear model neither collapses nor folds any infinitesimal direction. One expects the nonlinear map itself to be invertible in a sufficiently small neighborhood. The inverse function theorem turns this intuition into a theorem.

Theorem 4.5 (Inverse function theorem). *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 near a , and suppose $DF(a)$ is invertible. Then there are neighborhoods U of a and V of $F(a)$ such that $F : U \rightarrow V$ is a bijection with C^1 inverse. Moreover*

$$D(F^{-1})(F(x)) = [DF(x)]^{-1}.$$

Proof. Translate so $a = 0$ and $F(0) = 0$, and compose with $DF(0)^{-1}$, reducing to $DF(0) = I$. By continuity of DF , choose a closed ball $\overline{B}_r$ on which

$$\|DF(x) - I\| \leq q < 1.$$

For a target y , define

$$\Phi_y(x) = x - F(x) + y.$$

Then $D\Phi_y(x) = I - DF(x)$ has operator norm at most q . The mean-value estimate along line segments gives

$$\|\Phi_y(x) - \Phi_y(x')\| \leq q \|x - x'\|,$$

so Φ_y is a contraction. Also $\Phi_0(0) = 0$, hence for $\|x\| \leq r$,

$$\|\Phi_y(x)\| \leq \|\Phi_0(x) - \Phi_0(0)\| + \|y\| \leq qr + \|y\|.$$

Thus whenever $\|y\| \leq (1 - q)r$, the map Φ_y sends $\overline{B}_r$ into itself. By the contraction mapping theorem, it has a unique fixed point x , and the fixed-point equation is exactly $F(x) = y$. The same contraction estimate shows two points in the ball cannot have the same F -value. Thus F is locally one-to-one and onto a neighborhood of 0.

It remains to prove differentiability of the inverse. Let $y = F(x)$ and $y + k = F(x + h)$. Differentiability gives

$$k = DF(x)h + o(\|h\|).$$

The lower Lipschitz estimate coming from $\|DF - I\| \leq q < 1$ implies $\|h\| = O(\|k\|)$. Hence

$$h = [DF(x)]^{-1}k + o(\|k\|),$$

which proves $D(F^{-1})(y) = [DF(x)]^{-1}$. Continuity follows from continuity of matrix inversion on the invertible matrices. $\square$

Proof architecture

The decisive trick is to turn the equation $F(x) = y$ into a fixed-point problem. Invertibility of the derivative gives a contraction after normalizing the derivative to the identity. Local inversion is therefore a stability theorem for fixed points.

4.8 Implicit function theorem: solving equations for some variables

Theorem 4.6 (Implicit function theorem, basic form). *Let $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ be C^1 , write variables as (x, y) , and suppose $F(a, b) = 0$. If the $m \times m$ matrix $D_y F(a, b)$ is invertible, then near (a, b) the equation $F(x, y) = 0$ determines y uniquely as a C^1 function $y = g(x)$. Moreover*

$$Dg(a) = -[D_y F(a, b)]^{-1} D_x F(a, b).$$

Proof. Define $H(x, y) = (x, F(x, y))$. Its derivative at (a, b) has block form

$$DH(a, b) = \begin{pmatrix} I & 0 \\ D_x F & D_y F \end{pmatrix},$$

which is invertible because $D_y F$ is invertible. By the inverse function theorem, H has a local inverse. Holding the second output coordinate equal to 0 gives $(x, g(x)) = H^{-1}(x, 0)$ and therefore $F(x, g(x)) = 0$. Differentiate this identity:

$$D_x F + D_y F Dg = 0,$$

and solve for Dg . □

Deep intuition

The implicit function theorem is a local “permission to solve.” You may not have a formula for y in terms of x , but if the equation responds nondegenerately to changes in y , then nearby there really is a unique hidden function $y = g(x)$.

4.9 Recall drill

Recall drill

State differentiability without coordinates. Explain why the derivative is linear, why the chain rule becomes composition, why the gradient depends on an inner product, what the Hessian measures, and why invertibility of the derivative should imply local invertibility.

4.10 Exercises

Exercise 4.1. Show directly from the definition that a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable everywhere and $DT(a) = T$.

Exercise 4.2. For $f(x, y) = x^2 + 4xy + 5y^2$, classify the critical point at the origin by diagonalizing the Hessian. Interpret the answer geometrically.

Exercise 4.3. Use Lagrange multipliers to find the maximum and minimum of xy on the circle $x^2 + y^2 = 1$. Then solve the same problem by a change of variables and compare the viewpoints.

Exercise 4.4. Show that the map $F(x, y) = (e^x \cos y, e^x \sin y)$ is locally invertible everywhere. Why is it not globally injective?

Exercise 4.5. Use the implicit function theorem to show that near $(1, 0)$ the equation $x^2 + y^2 = 1$ determines x as a smooth function of y . Compute $x'(0)$ and $x''(0)$ from the equation.

Topology: continuity without coordinates or distance

Topology begins with a radical simplification: to discuss continuity, we do not need to know exact distances. We need only know which sets count as neighborhoods. Once that information is isolated, ideas from analysis become portable to spaces where no natural metric is visible.

5.1 Why open sets are the primitive language

In a metric space (X, d) , a set U is open when every $x \in U$ has some ball $B_r(x)$ contained in U . Notice that continuity can be characterized entirely in terms of open sets: $f : X \rightarrow Y$ is continuous exactly when inverse images of open sets are open.

Motivation

A metric records far more than continuity needs. It distinguishes whether two points are 1 unit or 100 units apart, but continuity only asks whether arbitrarily small neighborhoods are respected. Topology keeps the neighborhood structure and discards unnecessary numerical information.

Definition 5.1. A *topology* on a set X is a collection $\tau \subseteq \mathcal{P}(X)$ such that

1. $\emptyset, X \in \tau$;
2. arbitrary unions of sets in τ belong to τ ;
3. finite intersections of sets in τ belong to τ .

The pair (X, τ) is a topological space, and sets in τ are called open.

Deep intuition

An open set is a region in which every point has room to move a little without leaving the region. A topology is a consistent declaration of what “room to move locally” means. Arbitrary unions are allowed because local room remains local when regions are assembled; only finite intersections are required because infinitely many neighborhood conditions can

squeeze room down to nothing.

Four topologies to calibrate intuition

- The usual topology on $\mathbb{R}$ consists of unions of open intervals.
- The discrete topology $\mathcal{P}(X)$ declares every set open; every map out of a discrete space is continuous.
- The indiscrete topology $\{\emptyset, X\}$ has almost no open sets; every map into an indiscrete space is continuous.
- On $\mathbb{R}$, the lower-limit topology generated by intervals $[a, b)$ is finer than the usual topology. It has more open sets and therefore makes continuity into it harder, but continuity out of it easier.

5.2 Bases: specifying topology efficiently

Definition 5.2. A collection $\mathcal{B}$ of subsets of X is a *basis* for a topology if

1. for every $x \in X$, some $B \in \mathcal{B}$ contains x ;
2. if $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then some $B_3 \in \mathcal{B}$ satisfies $x \in B_3 \subseteq B_1 \cap B_2$.

The open sets are arbitrary unions of basis elements.

Proposition 5.3. *The collection of open balls in a metric space is a basis for its metric topology.*

Proof. Every x lies in, say, $B_1(x)$. If $x \in B_r(a) \cap B_s(b)$, let

$$\delta = \min\{r - d(a, x), s - d(b, x)\} > 0.$$

Then $B_\delta(x)$ lies in both balls by the triangle inequality. □

Connection

A basis in topology plays the same compression role as a basis in linear algebra, but the algebra is different. A vector-space basis builds every vector by finite linear combination; a topological basis builds every open set by arbitrary union.

5.3 Closed sets, closure, interior, and boundary

A set F is *closed* if $X \setminus F$ is open. For $A \subseteq X$:

- $\text{Int } A$ is the largest open set contained in A ;
- $\text{cl } A$ is the smallest closed set containing A ;
- $\partial A = \text{cl } A \setminus \text{Int } A$ is the boundary.

Proposition 5.4. *A point x lies in $\text{cl } A$ if and only if every open neighborhood of x intersects A .*

Proof. If some neighborhood U of x misses A , then $X \setminus U$ is closed, contains A , and misses x , so $x \notin \text{cl } A$. Conversely, if $x \notin \text{cl } A$, the open set $X \setminus \text{cl } A$ is a neighborhood of x disjoint from A . $\square$

Deep intuition

Closure adds all points that are impossible to separate from the set by an open neighborhood. Boundary consists of points for which every local inspection sees both the set and its complement. This is a purely topological notion of an “edge.”

5.4 Continuity: inverse images preserve observable neighborhoods

Definition 5.5. A map $f : X \rightarrow Y$ is continuous if $f^{-1}(U)$ is open in X for every open $U \subseteq Y$.

Why inverse images rather than images? Inverse images automatically preserve unions, intersections, and complements:

$$f^{-1}\left(\bigcup U_i\right) = \bigcup f^{-1}(U_i), \quad f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V).$$

That makes them perfectly matched to the topology axioms.

Theorem 5.6. *For metric spaces, the open-set definition of continuity is equivalent to the ε - δ definition.*

Proof. Assume the ε - δ condition. If $U \subseteq Y$ is open and $x \in f^{-1}(U)$, choose $\varepsilon > 0$ with $B_\varepsilon(f(x)) \subseteq U$. Continuity gives $\delta > 0$ such that $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \varepsilon$, so $B_\delta(x) \subseteq f^{-1}(U)$. Thus the inverse image is open.

Conversely, suppose inverse images of open sets are open. Given x and $\varepsilon > 0$, $f^{-1}(B_\varepsilon(f(x)))$ is an open neighborhood of x , hence contains some $B_\delta(x)$. That is exactly the ε - δ condition. $\square$

Proposition 5.7. *Compositions of continuous maps are continuous.*

Proof. For open U in the target,

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)),$$

which is open because $g^{-1}(U)$ is open and then its inverse image under f is open. $\square$

5.5 Homeomorphism: sameness from the topological viewpoint

Definition 5.8. A *homeomorphism* is a bijection $f : X \rightarrow Y$ such that both f and f^{-1} are continuous. Homeomorphic spaces are topologically equivalent.

Geometry discarded, topology retained

The open interval $(-1, 1)$ and the whole real line are homeomorphic via

$$f(x) = \frac{x}{1 - |x|}.$$

Distances and boundedness are not preserved. What is preserved is neighborhood structure. By contrast, $[0, 1]$ is not homeomorphic to $(0, 1)$: compactness will distinguish them.

5.6 Compactness: no escape through infinitely many local choices

Definition 5.9. A space X is *compact* if every open cover of X has a finite subcover.

Motivation

Many proofs in analysis make a local choice at every point: a radius, a bound, a neighborhood, a coordinate chart. Without compactness, those infinitely many local choices may have no global uniform control. Compactness is the mechanism that turns local information into finitely many pieces and therefore into a global bound.

Theorem 5.10. *The continuous image of a compact space is compact.*

Proof. Let $f : X \rightarrow Y$ be continuous with X compact, and let $\{U_i\}$ cover $f(X)$. Then $\{f^{-1}(U_i)\}$ is an open cover of X . Choose a finite subcover $f^{-1}(U_{i_1}), \dots, f^{-1}(U_{i_k})$. Then $U_{i_1}, \dots, U_{i_k}$ cover $f(X)$. $\square$

Corollary 5.11 (Extreme value theorem). *A continuous real-valued function on a compact space attains its maximum and minimum.*

Proof. The image $f(X)$ is compact in $\mathbb{R}$, hence closed and bounded. Its supremum and infimum belong to the closed set $f(X)$. $\square$

Theorem 5.12 (Compactness gives uniform continuity). *A continuous map from a compact metric space (X, d_X) to a metric space (Y, d_Y) is uniformly continuous.*

Proof. Fix $\varepsilon > 0$. For each $x \in X$, continuity gives $r_x > 0$ such that

$$d_X(x, z) < 2r_x \implies d_Y(f(x), f(z)) < \varepsilon/2.$$

The balls $B_{r_x}(x)$ cover X . Choose a finite subcover $B_{r_{x_j}}(x_j)$ and let $\delta = \min_j r_{x_j} > 0$. If $d_X(u, v) < \delta$, choose j with $u \in B_{r_{x_j}}(x_j)$. Then

$$d_X(v, x_j) \leq d_X(v, u) + d_X(u, x_j) < \delta + r_{x_j} \leq 2r_{x_j},$$

so both $f(u)$ and $f(v)$ are within $\varepsilon/2$ of $f(x_j)$. Hence $d_Y(f(u), f(v)) < \varepsilon$. $\square$

Deep intuition

Uniform continuity is continuity with one scale working everywhere. Compactness makes this possible because infinitely many pointwise scales can be reduced to finitely many, and a minimum of finitely many positive radii stays positive.

5.7 Sequential compactness in metric spaces

Theorem 5.13. *For metric spaces, compactness is equivalent to sequential compactness: every sequence has a convergent subsequence.*

Proof. Assume compactness. If a sequence (x_n) had no convergent subsequence, then for each $x \in X$ some neighborhood of x would contain only finitely many x_n ; otherwise one could choose indices increasing to infinity from nested balls around x and obtain a subsequence converging to x . These neighborhoods cover X . A finite subcover would contain only finitely many terms altogether, contradiction.

Conversely, assume sequential compactness and suppose an open cover $\mathcal{U}$ has no finite subcover. First show there is a Lebesgue number $\delta > 0$: some δ such that every ball of radius δ is contained in one member of $\mathcal{U}$. If not, for each n choose x_n such that $B_{1/n}(x_n)$ lies in no cover element. A convergent subsequence $x_{n_k} \rightarrow x$ lies eventually in some $U \in \mathcal{U}$ together with a small ball around it, contradicting the choice of x_{n_k} . Now sequential compactness also implies total boundedness: for every $\delta > 0$, finitely many δ -balls cover X ; otherwise choose successively points separated by at least δ , producing a sequence with no Cauchy, hence no convergent, subsequence. Cover X by finitely many balls of radius $\delta/2$; each lies in some member of $\mathcal{U}$, giving a finite subcover. $\square$

Common pitfall

In arbitrary topological spaces, compactness and sequential compactness need not coincide. Sequences are sufficient probes in metric spaces but can miss topological behavior in more general spaces. Nets and filters repair this at the graduate level.

5.8 Connectedness: spaces that cannot be split cleanly

Definition 5.14. A space X is connected if it cannot be written as $X = U \cup V$ with U, V disjoint, nonempty, open subsets of X .

Theorem 5.15. *The continuous image of a connected space is connected.*

Proof. Suppose $f(X) = A \cup B$ were a separation. Then $f^{-1}(A)$ and $f^{-1}(B)$ would be disjoint nonempty open subsets of X whose union is X , contradicting connectedness. $\square$

Corollary 5.16 (Intermediate value theorem). *If X is connected and $f : X \rightarrow \mathbb{R}$ is continuous, then $f(X)$ is an interval. In particular, a continuous function on $[a, b]$ takes every value between $f(a)$ and $f(b)$.*

Deep intuition

Connectedness is the topological form of “no jump can separate the space into two open camps.” The intermediate value theorem is therefore not fundamentally about formulas on intervals; it is about continuous images being unable to create a disconnection that was not present in the domain.

5.9 Product topology: observing finitely many coordinates at a time

For spaces X and Y , the product topology on $X \times Y$ has basis $U \times V$ with U open in X and V open in Y .

For arbitrary products $\prod_{i \in I} X_i$, a basic open set restricts only finitely many coordinates and leaves the rest unrestricted.

Deep intuition

Why only finitely many coordinates? Because a neighborhood should represent finite observational precision. In an infinite product, demanding an independent tiny restriction in every coordinate produces a topology that is too fine for many natural compactness and convergence results. The product topology says: a finite experiment can inspect only finitely many coordinates at once.

5.10 Quotient topology: glue first, then define openness

Let $q : X \rightarrow Y$ be a surjection. The quotient topology on Y declares $U \subseteq Y$ open exactly when $q^{-1}(U)$ is open in X .

Motivation

Quotients create new spaces by identifying points. The quotient topology is the strongest topology on the glued space making the identification map continuous. It therefore records exactly the open sets that can be detected upstairs before gluing.

Making a circle from an interval

Identify $0 \sim 1$ in $[0, 1]$. The quotient is homeomorphic to S^1 via $t \mapsto e^{2\pi it}$. A neighborhood of the glued point corresponds upstairs to a set containing small intervals near both 0 and 1. This explains why the endpoints become locally adjacent on the circle.

5.11 Hausdorff spaces: limits should be distinguishable

Definition 5.17. A space is Hausdorff if distinct points $x \neq y$ have disjoint neighborhoods.

Proposition 5.18. *Limits of sequences are unique in Hausdorff spaces.*

Proof. If $x_n \rightarrow x$ and $x_n \rightarrow y$ with $x \neq y$, choose disjoint neighborhoods U of x and V of y . Eventually $x_n \in U$ and eventually $x_n \in V$, impossible. $\square$

Proposition 5.19. *Compact subsets of Hausdorff spaces are closed.*

Proof. Let K be compact and $x \notin K$. For each $y \in K$, choose disjoint neighborhoods U_y of y and V_y of x . Finitely many $U_{y_1}, \dots, U_{y_n}$ cover K . Then $V = V_{y_1} \cap \dots \cap V_{y_n}$ is an open neighborhood of x disjoint from K . Hence $X \setminus K$ is open. $\square$

5.12 Homotopy: deforming maps continuously

Definition 5.20. Maps $f, g : X \rightarrow Y$ are *homotopic* if there is a continuous $H : X \times [0, 1] \rightarrow Y$ with $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. A space is *contractible* if its identity map is homotopic to a constant map.

Convex sets are contractible

If $C \subseteq \mathbb{R}^n$ is convex and $a \in C$, then

$$H(x, t) = (1 - t)x + ta$$

stays in C and continuously shrinks C to a . Thus every convex set is contractible. A circle is not, but proving this requires an invariant such as the fundamental group.

Deep intuition

A homeomorphism demands exact reversible topological equivalence. Homotopy is coarser: it asks whether one map can flow continuously into another. Algebraic topology succeeds by attaching algebraic invariants to spaces that remain unchanged under homotopy. Holes become measurable because algebra cannot be continuously deformed away.

5.13 Fundamental group: loops as algebraic probes of holes

Fix $x_0 \in X$. A loop is a continuous $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = \gamma(1) = x_0$. Two loops are equivalent if they are homotopic through loops keeping the basepoint fixed. The equivalence classes form the fundamental group $\pi_1(X, x_0)$ under concatenation.

Proposition 5.21. *Loop concatenation is well-defined on homotopy classes and gives a group.*

Proof. If $\gamma_0 \simeq \gamma_1$ and $\eta_0 \simeq \eta_1$ through based homotopies, concatenate the homotopies in the path parameter to obtain $\gamma_0 * \eta_0 \simeq \gamma_1 * \eta_1$. The constant loop is an identity up to reparametrization homotopy; reversing a loop gives an inverse because traversing a path and retracing it contracts to the basepoint. Associativity also holds up to a reparametrization homotopy, so it holds exactly on homotopy classes. $\square$

Connection

The slogan $\pi_1(S^1) \cong \mathbb{Z}$ says a loop around a circle is classified by its winding number. Establishing this rigorously uses the covering map $\mathbb{R} \rightarrow S^1$, $t \mapsto e^{2\pi it}$, and path lifting. The integer records how far a lifted loop travels upstairs. This is the prototype of converting a geometric hole into algebra.

5.14 Recall drill**Recall drill**

Explain why arbitrary unions but only finite intersections appear in the topology axioms; why continuity uses inverse images; how compactness converts local data into finite data; how connectedness underlies the intermediate value theorem; what quotient topology does after points are glued; and what a fundamental group is trying to detect.

5.15 Exercises

Exercise 5.1. Show that the topology generated by the basis of half-open intervals $[a, b)$ on $\mathbb{R}$ is strictly finer than the usual topology.

Exercise 5.2. Prove that a subset A of a metric space is closed if and only if every convergent sequence in A has its limit in A .

Exercise 5.3. Prove that a continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Exercise 5.4. Show that the product of two connected spaces is connected. Hint: connect each (x, y) to a fixed vertical and horizontal slice.

Exercise 5.5. Let $q : [0, 1] \rightarrow S^1$ be $q(t) = e^{2\pi it}$. Prove directly from the quotient topology that the induced bijection $[0, 1]/(0 \sim 1) \rightarrow S^1$ is a homeomorphism.

Exercise 5.6. Show that $\mathbb{R}^n$ is path-connected and contractible. Show that a disconnected space cannot be path-connected.

Discrete mathematics

Discrete mathematics is sometimes presented as a bag of clever tricks. A better view is that it trains a small number of proof mechanisms unusually intensely: encode objects, count the same set in two ways, recurse on size, preserve an invariant, choose an extremal object, or turn local restrictions into global structure.

6.1 The basic counting principles

If a task consists of disjoint alternatives with $a_1, \dots, a_k$ possibilities, the total is $a_1 + \dots + a_k$. If it consists of successive independent choices with $a_1, \dots, a_k$ possibilities, the total is $a_1 \cdots a_k$.

Deep intuition

Addition corresponds to partitioning a set into disjoint cases. Multiplication corresponds to identifying an object with a tuple of independent choices. Good counting proofs usually begin by making one of these structural decompositions explicit.

Strings with restrictions

How many binary strings of length n contain exactly k ones? Choose the k positions of the ones: $\binom{n}{k}$. How many contain at least one one? Either sum $\sum_{k=1}^n \binom{n}{k}$ or use the complement: $2^n - 1$. The second method is shorter because the forbidden case is structurally simpler.

6.2 Bijections: proving two counts are equal without counting either directly

Proposition 6.1 (Symmetry of binomial coefficients).

$$\binom{n}{k} = \binom{n}{n-k}.$$

Proof. Map each k -element subset $A \subseteq \{1, \dots, n\}$ to its complement. This is a bijection from k -subsets to $(n - k)$ -subsets. $\square$

Proposition 6.2 (Pascal identity).

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Proof. Count k -subsets of $\{1, \dots, n\}$ by whether they contain n . Those that do not are k -subsets of the first $n-1$ elements; those that do correspond to $(k-1)$ -subsets after removing n . $\square$

Motivation

A bijection is stronger than an equality of numbers: it explains why the two families are the same size by pairing their members. Many mysterious algebraic identities become obvious after the right combinatorial objects are found.

6.3 Pigeonhole principle: compression forces collision

Theorem 6.3 (Pigeonhole principle). *If N objects are placed in k boxes, some box contains at least $\lceil N/k \rceil$ objects.*

Proof. If every box contained at most $\lceil N/k \rceil - 1$ objects, the total would be at most

$$k(\lceil N/k \rceil - 1) < N,$$

a contradiction. $\square$

Deep intuition

Pigeonhole is an information theorem: a map from a larger finite set into a smaller one cannot be injective. Whenever many objects are classified by relatively few possible “signatures,” some two objects must share a signature. The art is choosing the signature.

Two integers with divisible difference

Among any $n+1$ integers, two have difference divisible by n . Classify each integer by its residue modulo n . There are only n residue classes, so two integers have the same class.

6.4 Inclusion-exclusion: correcting overcount systematically

For finite sets $A_1, \dots, A_n$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots.$$

Theorem 6.4 (Inclusion-exclusion). *The formula above holds for finite sets.*

Proof. Fix an element x belonging to exactly $r \geq 1$ of the sets. Its total contribution to the right side is

$$\binom{r}{1} - \binom{r}{2} + \cdots + (-1)^{r+1} \binom{r}{r} = 1 - (1-1)^r = 1.$$

An element belonging to none contributes 0. Thus every element of the union is counted exactly once. $\square$

Counting surjections

The number of onto functions from an n -element set to a k -element set is

$$\sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

Start with all k^n functions and, for each target value i , let A_i be the functions that miss i . Inclusion-exclusion corrects for functions missing several target values.

6.5 Recurrences: build larger objects from smaller ones

A recurrence does not merely provide a formula; it expresses structural self-similarity.

Fibonacci from tilings

Let a_n count tilings of a length- n board by squares of length 1 and dominoes of length 2. Inspect the first tile. If it is a square, a_{n-1} choices remain; if a domino, a_{n-2} remain. Hence

$$a_n = a_{n-1} + a_{n-2}, \quad a_0 = a_1 = 1.$$

The recurrence is a partition of objects by their first local decision.

For a linear recurrence $a_n = \alpha a_{n-1} + \beta a_{n-2}$, exponential sequences r^n are natural because shifting n only multiplies by r . The same idea extends to

$$a_n = \alpha a_{n-1} + \beta a_{n-2} :$$

try $a_n = r^n$ and obtain the characteristic equation $r^2 = \alpha r + \beta$.

Connection

This is eigenvalue thinking in disguise. A recurrence advances a state vector, for example

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix}.$$

Solving the recurrence means understanding powers of a linear operator.

6.6 Strong induction and minimal counterexamples

Strong induction assumes $P(1), \dots, P(n)$ to prove $P(n+1)$. It is equivalent to ordinary induction, but often mirrors the recursive structure more naturally.

Theorem 6.5 (Fundamental theorem of arithmetic: existence). *Every integer $n \geq 2$ is a product of primes.*

Proof. Use strong induction. If n is prime, we are done. If not, $n = ab$ with $2 \leq a, b < n$. By the induction hypothesis, both a and b are products of primes, hence so is n . $\square$

Deep intuition

The minimal-counterexample method is induction read backwards. If a theorem failed, choose the smallest failure. Any operation that turns it into a smaller object produces something that must satisfy the theorem, and that information is used to repair the alleged failure.

6.7 Invariants and monovariants

An invariant is a quantity unchanged by allowed moves. A monovariant changes only in one direction and is often used to prove termination.

Checkerboard coloring

Remove two opposite corners from an 8×8 chessboard. Can the remaining board be tiled by 1×2 dominoes? Color the board black and white. Opposite corners have the same color, so after removal there are 30 squares of one color and 32 of the other. Every domino covers one black and one white square. The color difference is an invariant obstruction, so tiling is impossible.

Euclidean algorithm as a monovariant

In the Euclidean algorithm, replacing (a, b) by $(b, a \bmod b)$ strictly decreases the second positive coordinate. Since positive integers cannot decrease forever, the algorithm terminates. The decreasing remainder is a monovariant.

6.8 Equivalence relations and partitions

Definition 6.6. A relation $\sim$ on X is an equivalence relation if it is reflexive, symmetric, and transitive. The equivalence class of x is $[x] = \{y : y \sim x\}$.

Theorem 6.7. *Equivalence relations on X are in bijection with partitions of X .*

Proof. Given an equivalence relation, classes are nonempty, cover X , and two classes are either equal or disjoint: if $[x] \cap [y] \neq \emptyset$, choose z in the intersection; then $x \sim z \sim y$, so $x \sim y$, and

transitivity shows the classes coincide. Conversely, given a partition, declare $x \sim y$ when they lie in the same block. Reflexivity, symmetry, and transitivity are immediate. $\square$

Deep intuition

An equivalence relation is a rule for forgetting distinctions. A partition is the resulting geometry of indistinguishability. Quotient groups, quotient spaces, congruence classes, and connected components all follow this pattern.

6.9 Partial orders: structure without total comparability

Definition 6.8. A partial order $\leq$ is reflexive, antisymmetric, and transitive. A poset need not make every pair comparable.

Examples include subset inclusion on $\mathcal{P}(X)$ and divisibility on positive integers. A Hasse diagram suppresses reflexive and transitive edges to reveal the immediate-order structure.

Theorem 6.9 (Finite posets have maximal elements). *Every nonempty finite poset has a maximal element.*

Proof. Start at any element x_0 . If it is not maximal, choose $x_1 > x_0$. Continue. A strictly increasing chain cannot revisit an element and the poset is finite, so the process stops at a maximal element. $\square$

Common pitfall

Maximal and maximum are different. A maximum is above every element and is unique if it exists. A maximal element merely has nothing strictly above it; there can be many incomparable maximal elements.

6.10 Generating functions: turn recurrence into algebra

Given a sequence $(a_n)_{n \geq 0}$, its ordinary generating function is the formal power series

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

We may use it formally, without asking whether the series converges analytically.

Fibonacci generating function

For $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$, set $F(x) = \sum_{n \geq 0} F_n x^n$. Then

$$F(x) - x = xF(x) + x^2 F(x),$$

so

$$F(x) = \frac{x}{1 - x - x^2}.$$

Factoring the denominator and expanding by partial fractions recovers Binet's formula. A shift recurrence has become an algebraic equation.

Deep intuition

A generating function is a container in which the index n becomes an exponent. Operations on indices become algebraic operations on power series: shifting corresponds to multiplication by x , convolution corresponds to multiplication of series, and weighted sums often correspond to differentiation.

6.11 Recall drill

Recall drill

Explain counting by partition versus counting by product; why bijections are explanatory; pigeonhole as failure of injectivity; inclusion-exclusion as elementwise correction; recurrence as self-similarity; invariant versus monovariant; equivalence relation as deliberate forgetting; and generating functions as index-to-exponent encoding.

6.12 Exercises

Exercise 6.1. Prove Vandermonde's identity

$$\binom{m+n}{r} = \sum_k \binom{m}{k} \binom{n}{r-k}$$

by counting subsets in two ways.

Exercise 6.2. How many integers in $\{1, \dots, 1000\}$ are divisible by 2, 3, or 5? Use inclusion-exclusion.

Exercise 6.3. Prove that among any $n^2 + 1$ distinct real numbers, there is an increasing subsequence of length $n + 1$ or a decreasing subsequence of length $n + 1$. Hint: assign each term two lengths.

Exercise 6.4. Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 2, a_1 = 5$. Then express the recurrence as a matrix iteration.

Exercise 6.5. Let a_n be the number of binary strings of length n with no consecutive 1s. Derive a recurrence and a generating function.

Exercise 6.6. Prove that every finite poset has a minimal element and a maximal element. Give an example with several of each but no minimum and no maximum.

Graph theory: from local relations to global structure

A graph strips a system down to objects and pairwise relationships. That severe simplification is why graphs appear everywhere: networks, algorithms, chemistry, social systems, scheduling, topology, and combinatorics. The central tension is local versus global: what can local degree information, forbidden configurations, or path rules force about the whole graph?

7.1 Basic language

A finite simple graph is $G = (V, E)$, where E is a collection of two-element subsets of V . The degree $\deg(v)$ is the number of incident edges. A path is a sequence of adjacent vertices; a cycle is a closed path with no repeated vertices except the endpoints.

Theorem 7.1 (Handshake lemma). *For every finite graph,*

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. Count incidences (v, e) with vertex v incident to edge e . Summing by vertices gives the left side. Summing by edges gives $2|E|$, because each edge has two endpoints. $\square$

Corollary 7.2. *Every finite graph has an even number of odd-degree vertices.*

Deep intuition

The handshake lemma is double counting in its purest form. One set of incidences is viewed from two coordinate systems: “which vertex owns this incidence?” versus “which edge owns it?” Many graph identities are obtained by inventing the right incidence set.

7.2 Connectivity and components

Define $u \sim v$ if there is a path from u to v . This is an equivalence relation; its classes are the connected components.

Proof. Reflexivity uses the length-zero path, symmetry reverses a path, and transitivity concatenates paths. $\square$

Connection

Connected components are a quotient construction: vertices are grouped according to the relation “can reach each other.” This parallels path components in topology and communicating classes in Markov chains.

7.3 Trees: connectedness with no redundancy

Definition 7.3. A tree is a connected graph with no cycles.

Theorem 7.4 (Equivalent characterizations of trees). *For a finite graph G with n vertices, the following are equivalent:*

1. G is a tree;
2. there is a unique path between every pair of vertices;
3. G is connected and has $n - 1$ edges;
4. G has no cycles and has $n - 1$ edges;
5. G is connected and every edge is a bridge: deleting any edge disconnects the graph.

Proof. (1) $\Rightarrow$ (2): connectedness gives a path. Two distinct simple paths between the same vertices would combine to contain a cycle.

(2) $\Rightarrow$ (5): if deleting an edge uv did not disconnect u from v , another u - v path would exist, contradicting uniqueness.

(5) $\Rightarrow$ (1): connectedness is part of (5). If a cycle existed, deleting an edge of the cycle would leave its endpoints connected through the rest of the cycle, so the graph would remain connected, contradicting the bridge condition.”

For the edge-count characterizations, first prove by induction that every tree with $n \geq 2$ has a leaf: take a longest simple path; an endpoint cannot have a neighbor outside the path or a second neighbor on the path without extending the path or creating a cycle. Remove a leaf and its incident edge. The remainder is a tree, so induction gives $n - 2$ edges there and $n - 1$ in total. Thus (1) $\Rightarrow$ (3).

If a connected graph has $n - 1$ edges and contains a cycle, delete an edge from the cycle; connectivity remains, yielding a connected graph with $n - 2$ edges. But any connected n -vertex graph has at least $n - 1$ edges: grow a connected set from one vertex, and each new vertex requires a distinct edge. Contradiction. Hence (3) $\Rightarrow$ (1).

If an acyclic graph with n vertices has $n - 1$ edges and had $c > 1$ components, each component is a tree, so the total number of edges would be $n - c < n - 1$. Thus (4) $\Rightarrow$ (1). The remaining implications follow from the established equivalences. $\square$

Deep intuition

A tree is exactly connectedness with no spare edges. Every edge carries essential global information. Adding one edge creates exactly one cycle; deleting one edge creates exactly two components. Trees sit on the knife-edge between disconnectedness and redundancy.

7.4 Spanning trees and minimal connectivity

Every finite connected graph contains a spanning tree.

Proof. Start with the connected graph. While a cycle exists, delete one edge from that cycle. Deleting a cycle edge preserves connectivity. The process terminates because the finite edge count decreases. The final spanning subgraph is connected and acyclic, hence a tree. $\square$

This proves that $n - 1$ edges are enough to encode the connectivity skeleton of any connected n -vertex graph.

7.5 Euler trails: using every edge exactly once

Theorem 7.5 (Euler criterion). *A finite connected graph has an Euler circuit if and only if every vertex has even degree. It has an Euler trail with distinct endpoints if and only if exactly two vertices have odd degree.*

Proof. Necessity: each time a trail enters an internal vertex by an unused edge, it must leave by another, pairing incident edges. In a circuit this applies to every vertex; in an open trail only the start and end have one unpaired incident edge.

For sufficiency in the all-even case, begin at any vertex and follow unused edges until no continuation is possible. Because every intermediate visit uses edges in enter/leave pairs, one cannot get stuck away from the starting vertex. Thus a closed trail C is obtained. If it uses all edges, we are done. Otherwise, connectedness gives a vertex of C incident to an unused edge. Starting there, build another closed trail from unused edges and splice it into C . Repeating finitely many times yields an Euler circuit. If exactly two vertices u, v are odd, add an extra edge uv , find an Euler circuit, and delete the added edge to obtain an Euler trail from u to v . $\square$

7.6 Bipartite graphs and odd cycles

Theorem 7.6. *A graph is bipartite if and only if it contains no odd cycle.*

Proof. If G is bipartite, every cycle alternates between the two parts and therefore has even length.

Conversely, assume no odd cycle. In each connected component choose a root r . Put a vertex in part A if its distance from r is even and in B if odd. If an edge joined two vertices of the same parity, shortest paths from them to r together with the edge would contain an odd cycle (remove

their common initial segment from the root to obtain two internally disjoint tails whose lengths have the same parity, plus one edge). Therefore every edge crosses between A and B . $\square$

Deep intuition

Bipartiteness is a global two-coloring condition, yet it is exactly obstructed by one local-looking pattern: an odd cycle. This is a model example of graph theory turning a global feasibility problem into a forbidden-structure theorem.

7.7 Matchings and Hall's theorem

A matching is a set of pairwise vertex-disjoint edges. In a bipartite graph with parts L, R , a matching *saturates* L if every vertex of L is matched.

For $S \subseteq L$, let $N(S) \subseteq R$ be its neighborhood.

Theorem 7.7 (Hall's marriage theorem). *A finite bipartite graph has a matching saturating L if and only if*

$$|N(S)| \geq |S| \quad \text{for every } S \subseteq L.$$

Proof. Necessity is immediate: distinct matched vertices of S require distinct neighbors in $N(S)$.

For sufficiency, use induction on $|L|$. If every nonempty proper $S \subsetneq L$ satisfies the strict inequality $|N(S)| \geq |S| + 1$, choose any edge ℓr . Delete ℓ and r . For a subset S of the remaining left vertices, its new neighborhood has size at least $|N(S)| - 1 \geq |S|$, so Hall holds and induction gives a matching of the remainder; add ℓr .

Otherwise there is a nonempty proper $S \subsetneq L$ with $|N(S)| = |S|$. Hall holds for the induced bipartite graph on $S \cup N(S)$, so induction matches S to $N(S)$. Consider the remaining graph with left side $L \setminus S$ and right side $R \setminus N(S)$. For $T \subseteq L \setminus S$, Hall applied to $S \cup T$ gives

$$|N(S \cup T)| \geq |S| + |T|.$$

At most $|N(S)| = |S|$ of these neighbors lie in $N(S)$, so at least $|T|$ lie outside. Hence Hall holds for the remainder, which can also be matched by induction. Combining the two matchings saturates L . $\square$

Motivation

Hall's condition says every group of applicants collectively has enough possible jobs. The theorem is surprising because these exponentially many local capacity inequalities are not merely necessary; together they are sufficient for a global assignment.

7.8 Coloring and greedy bounds

A proper vertex coloring assigns colors so adjacent vertices receive different colors. The least number is the chromatic number $\chi(G)$.

Proposition 7.8. *If the maximum degree is Δ , then $\chi(G) \leq \Delta + 1$.*

Proof. Order the vertices arbitrarily and color them one at a time. When coloring a vertex, at most Δ colors are forbidden by already colored neighbors, so among $\Delta + 1$ colors at least one remains. $\square$

Common pitfall

The greedy algorithm proves an upper bound but need not find an optimal coloring. The outcome can depend dramatically on vertex order. This is common in algorithms: a constructive proof may certify feasibility without solving the optimization problem.

7.9 Planarity and Euler's formula

For a connected planar graph embedded without crossings, let V, E, F denote the numbers of vertices, edges, and faces, including the outer face.

Theorem 7.9 (Euler formula).

$$V - E + F = 2.$$

Proof. If the graph is a tree, $E = V - 1$ and $F = 1$, so the formula holds. If the graph contains a cycle, delete an edge on a cycle. The graph remains connected, E decreases by 1, and two adjacent faces merge so F also decreases by 1. Thus $V - E + F$ is unchanged. Repeat until a tree remains. $\square$

Corollary 7.10. *A simple connected planar graph with $V \geq 3$ satisfies $E \leq 3V - 6$.*

Proof. Every face boundary has length at least 3, and each edge borders two face sides, so $3F \leq 2E$. Euler gives

$$2 = V - E + F \leq V - E + \frac{2E}{3} = V - \frac{E}{3}.$$

Thus $E \leq 3V - 6$. $\square$

7.10 Recall drill

Recall drill

Explain the handshake lemma as double counting; why trees have exactly $n - 1$ edges; the parity mechanism behind Euler trails; why odd cycles obstruct bipartiteness; what Hall's condition measures; and how Euler's planar formula is preserved when cycle edges are removed.

7.11 Exercises

Exercise 7.1. Prove that every finite tree with at least two vertices has at least two leaves.

Exercise 7.2. Show that adding one edge to a tree creates exactly one cycle.

Exercise 7.3. Determine whether the complete bipartite graph $K_{3,3}$ is planar using Euler's formula and the fact that every face in a bipartite planar graph has boundary length at least 4.

Exercise 7.4. Prove that every graph has a bipartite subgraph containing at least half of its edges. Hint: choose a bipartition maximizing crossing edges.

Exercise 7.5. Use Hall's theorem to prove that every k -regular bipartite graph with $k \geq 1$ has a perfect matching.

Exercise 7.6. A connected graph has exactly two vertices of odd degree. Prove that every edge lies in some trail from one odd vertex to the other if and only if the graph has an Euler trail. Clarify the distinction between “some trail” and one trail using all edges.

Elementary number theory: divisibility, congruence, and finite arithmetic

Number theory is the study of the integers as a rigid multiplicative world. Its elementary tools look simple, but they train themes that recur throughout algebra: division with remainder, ideals generated by gcds, quotient rings, cyclic groups, local information modulo primes, and the passage from additive to multiplicative structure.

8.1 Divisibility and the Euclidean algorithm

For integers a, b , write $a \mid b$ if $b = ak$ for some integer k . A common divisor of a, b divides their greatest common divisor $\gcd(a, b)$.

Theorem 8.1 (Division algorithm). *For integers a and $b > 0$, there exist unique integers q, r such that*

$$a = bq + r, \quad 0 \leq r < b.$$

Proof. Consider the set $S = \{a - bq : q \in \mathbb{Z}, a - bq \geq 0\}$. It is nonempty for sufficiently negative q . By well-ordering, it has a least element $r = a - bq$. If $r \geq b$, then $r - b = a - b(q + 1)$ is a smaller nonnegative element of S , contradiction. Thus $0 \leq r < b$. If $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$, then $b(q - q') = r' - r$, whose absolute value is $< b$; the only multiple of b with that property is 0, so $q = q'$ and $r = r'$. $\square$

The Euclidean algorithm repeatedly applies

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Proposition 8.2. *If $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.*

Proof. A number divides both a and b exactly when it divides both b and $a - bq = r$. Thus the common divisors are identical. $\square$

Deep intuition

The Euclidean algorithm is a complexity-reduction machine. It preserves the invariant we care about - the set of common divisors - while replacing the pair by a strictly smaller one. Good algorithms often have exactly this architecture: preserve meaning, decrease size.

8.2 Bezout's identity and gcd as a generated ideal

Theorem 8.3 (Bezout). *For integers a, b , not both zero, there exist $x, y \in \mathbb{Z}$ such that*

$$ax + by = \gcd(a, b).$$

Proof. Let $S = \{ax + by > 0 : x, y \in \mathbb{Z}\}$. Choose its least element $d = ax_0 + by_0$. Divide $a = qd + r$ with $0 \leq r < d$. Then

$$r = a - q(ax_0 + by_0) = a(1 - qx_0) + b(-qy_0)$$

is another integer combination. Minimality forces $r = 0$, so $d \mid a$; similarly $d \mid b$. Every common divisor of a, b divides every integer combination, hence divides d . Therefore $d = \gcd(a, b)$. $\square$

Connection

The set of all integer combinations $ax + by$ is the ideal $(a, b) \subseteq \mathbb{Z}$. Bezout says

$$(a, b) = (\gcd(a, b)).$$

So the gcd is not merely a numerical statistic; it is a generator of the smallest ideal containing a and b .

8.3 Prime factorization and Euclid's lemma

Lemma 8.4 (Euclid). *If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$. Bezout gives $px + ay = 1$. Multiply by b :

$$pbx + aby = b.$$

Both terms on the left are divisible by p , so $p \mid b$. $\square$

Theorem 8.5 (Fundamental theorem of arithmetic). *Every integer $n \geq 2$ can be written as a product of primes, uniquely up to order.*

Proof. Existence was proved by strong induction in Chapter 5. For uniqueness, suppose

$$p_1 \cdots p_r = q_1 \cdots q_s.$$

Then p_1 divides the right side, so by repeated Euclid's lemma it divides some q_j . Since both are prime, $p_1 = q_j$. Cancel and continue inductively. $\square$

8.4 Congruence as equality in a quotient

Definition 8.6. For $n \geq 1$,

$$a \equiv b \pmod{n}$$

means $n \mid (a - b)$. This is an equivalence relation, and the classes form the ring $\mathbb{Z}/n\mathbb{Z}$.

Deep intuition

Congruence modulo n deliberately forgets how many full turns of size n separate two integers. Arithmetic survives because the equivalence relation is compatible with addition and multiplication. Modular arithmetic is therefore not a collection of remainder tricks; it is arithmetic in a quotient ring.

8.5 Linear congruences

Theorem 8.7. *The congruence*

$$ax \equiv b \pmod{n}$$

has a solution if and only if $d = \gcd(a, n)$ divides b . When solvable, it has exactly d incongruent solutions modulo n .

Proof. The congruence is equivalent to $ax + ny = b$ for some integer y . The set of all integers $ax + ny$ is precisely the multiples of d by Bezout, so solvability is equivalent to $d \mid b$. If $d \mid b$, divide by d to get

$$(a/d)x \equiv b/d \pmod{n/d}$$

with $\gcd(a/d, n/d) = 1$, so a/d has a multiplicative inverse modulo n/d , giving one class modulo n/d . This lifts to exactly d classes modulo n . $\square$

An inverse via the Euclidean algorithm

Solve $17x \equiv 5 \pmod{43}$. The Euclidean algorithm gives

$$43 = 2 \cdot 17 + 9, \quad 17 = 1 \cdot 9 + 8, \quad 9 = 1 \cdot 8 + 1.$$

Back-substitute:

$$1 = 9 - 8 = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Thus $17^{-1} \equiv -5 \equiv 38 \pmod{43}$, so $x \equiv 38 \cdot 5 \equiv 18 \pmod{43}$.

8.6 Chinese remainder theorem

Theorem 8.8 (CRT, integer form). *If m, n are coprime, then for any a, b there is a unique class modulo mn satisfying*

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}.$$

Equivalently,

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

Proof. Consider $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, $x \mapsto (x \bmod m, x \bmod n)$. Its kernel is the integers divisible by both m and n , which because $\gcd(m, n) = 1$ is $mn\mathbb{Z}$. For surjectivity, Bezout gives $um + vn = 1$. Then

$$x = b(um) + a(vn)$$

satisfies $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$. The first isomorphism theorem yields the ring isomorphism and uniqueness modulo mn . $\square$

Deep intuition

CRT says coprime moduli carry independent information. Knowing an integer modulo m and modulo n is equivalent to knowing it modulo mn . Arithmetic decomposes into smaller local clocks that can be recombined without interference.

8.7 Euler's totient and the multiplicative group of units

The units modulo n are residue classes $[a]$ with $\gcd(a, n) = 1$. They form the group $(\mathbb{Z}/n\mathbb{Z})^\times$. Its order is Euler's totient $\varphi(n)$.

If $n = \prod p_i^{e_i}$, then

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Indeed, for p^e , exactly p^{e-1} residues are divisible by p , so $\varphi(p^e) = p^e - p^{e-1}$. CRT makes φ multiplicative on coprime factors.

Theorem 8.9 (Euler's theorem). *If $\gcd(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. The residue class of a lies in the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$. Lagrange's theorem implies the order of a divides $\varphi(n)$, hence $a^{\varphi(n)} = 1$ in the group. $\square$

Corollary 8.10 (Fermat's little theorem). *If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.*

8.8 RSA as a structural example

Choose distinct primes p, q , let $n = pq$, and choose e coprime to $\varphi(n)$. Choose d with $ed \equiv 1 \pmod{\varphi(n)}$. Encryption maps m to $m^e \pmod{n}$, decryption raises to the power d .

For m coprime to n ,

$$m^{ed} = m^{1+k\varphi(n)} \equiv m \pmod{n}$$

by Euler. The statement can be extended to all residues modulo n by checking modulo p and q separately and applying CRT.

Deep intuition

RSA is a good synthesis example because the forward operation is easy - modular exponentiation - while constructing the inverse exponent uses hidden multiplicative structure. The mathematics is not “big numbers are hard”; it is the asymmetry between knowing n and knowing its factorization, which reveals $\varphi(n)$.

8.9 Quadratic residues: asking whether a square root exists modulo a prime

For an odd prime p , a nonzero residue a is a quadratic residue if $x^2 \equiv a \pmod{p}$ has a solution.

Lemma 8.11 (Root bound over a field). *A nonzero polynomial of degree d over a field has at most d roots.*

Proof. Induct on d . The claim is clear for $d = 0$. If f has a root a , the factor theorem gives $f(x) = (x - a)g(x)$ with $\deg g = d - 1$. Every root of f other than a is a root of g , because fields have no zero divisors. By induction there are at most $d - 1$ such roots. If f has no root, the bound is automatic. $\square$

Proposition 8.12 (Euler criterion). *For $p \nmid a$,*

$$a^{(p-1)/2} \equiv \begin{cases} 1 & \pmod{p}, & a \text{ is a quadratic residue,} \\ -1 & \pmod{p}, & a \text{ is a nonresidue.} \end{cases}$$

Proof. If $a \equiv b^2 \pmod{p}$, then Fermat’s little theorem gives

$$a^{(p-1)/2} \equiv b^{p-1} \equiv 1 \pmod{p}.$$

There are exactly $(p-1)/2$ nonzero quadratic residues: among the $p-1$ nonzero classes, the equation $x^2 = y^2$ is equivalent to $(x-y)(x+y) = 0$ in the field $\mathbb{Z}/p\mathbb{Z}$, hence $x = \pm y$, so squaring is exactly two-to-one.

The polynomial $X^{(p-1)/2} - 1$ over the field $\mathbb{Z}/p\mathbb{Z}$ has degree $(p-1)/2$, so it has at most that many roots. We have just exhibited $(p-1)/2$ roots, namely all nonzero squares; therefore those are exactly its roots. Finally Fermat gives

$$\left(a^{(p-1)/2}\right)^2 = a^{p-1} \equiv 1 \pmod{p}.$$

Since a field has no zero divisors, $z^2 = 1$ implies $z = \pm 1$. A nonresidue is not a root of $X^{(p-1)/2} - 1$, so its value must be -1 . $\square$

Deep intuition

Euler’s criterion turns an existence question - “does $x^2 = a$ have a solution?” - into a single exponentiation test. The proof is a root-counting argument: exactly half of the nonzero field elements are squares, and a polynomial of the matching degree detects precisely that half.

8.10 Recall drill

Recall drill

Explain the Euclidean algorithm as invariant plus descent; Bezout as a statement about generated ideals; why Euclid's lemma is the engine of unique factorization; congruence as quotient equality; CRT as independence of coprime clocks; Euler's theorem as Lagrange's theorem in a finite unit group.

8.11 Exercises

Exercise 8.1. Use the Euclidean algorithm to compute $\gcd(252, 198)$ and express it as an integer combination of 252 and 198.

Exercise 8.2. Determine all solutions of $18x \equiv 30 \pmod{42}$.

Exercise 8.3. Solve simultaneously $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$, $x \equiv 4 \pmod{9}$.

Exercise 8.4. Prove that if p is prime, then $\binom{p}{k}$ is divisible by p for $1 \leq k \leq p-1$. Deduce $(a+b)^p \equiv a^p + b^p \pmod{p}$.

Exercise 8.5. Prove that there are infinitely many primes congruent to 3 (mod 4) by adapting Euclid's argument.

Exercise 8.6. For $p = 11$, determine all nonzero quadratic residues and check Euler's criterion on one residue and one nonresidue.

Probability: rigorous reasoning about uncertainty

Probability becomes conceptually clean when one separates three layers: a sample space of possible outcomes, events that can be asked about, and a probability measure that assigns consistent weights. Random variables then become functions on that space, and expectation becomes integration against those weights.

9.1 Probability spaces and events

Definition 9.1. A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is the sample space, $\mathcal{F}$ is a sigma-algebra of events, and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfies $\mathbb{P}(\Omega) = 1$ and countable additivity on pairwise disjoint events.

For finite or countable elementary models, one often takes $\mathcal{F} = \mathcal{P}(\Omega)$. For continuous models, restricting to a sigma-algebra is essential; Chapter 9 explains why.

Deep intuition

Probability is not fundamentally about “random numbers.” It is a calculus of incomplete information. An event is a yes/no question about the underlying outcome, and probability measures how much of the possible-world space is compatible with that event.

Proposition 9.2. *For events A, B ,*

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. Since $\Omega = A \sqcup A^c$, countable additivity gives the first formula. For the second, decompose

$$A \cup B = A \sqcup (B \setminus A), \quad B = (A \cap B) \sqcup (B \setminus A),$$

and eliminate $\mathbb{P}(B \setminus A)$. □

9.2 Conditional probability: change the universe of discourse

Definition 9.3. If $\mathbb{P}(B) > 0$, define

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Deep intuition

Conditioning does not alter which underlying world occurred. It changes which worlds remain under consideration. After learning B , the event B becomes the new universe and probabilities are renormalized inside it.

The multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$$

is simply the definition rearranged.

Theorem 9.4 (Law of total probability). *If $B_1, \dots, B_n$ form a partition of Ω with positive probabilities, then*

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Proof. The sets $A \cap B_i$ are disjoint and their union is A . Therefore

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

□

Theorem 9.5 (Bayes' formula). *Under the same hypotheses,*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

Proof. Use the multiplication rule in the numerator and the law of total probability in the denominator. □

Base rates matter

Suppose a condition has prevalence 1%. A test has sensitivity 99% and false-positive rate 5%. Then

$$\mathbb{P}(\text{condition} \mid +) = \frac{0.99(0.01)}{0.99(0.01) + 0.05(0.99)} \approx 0.167.$$

A highly sensitive test can still yield a surprisingly modest posterior probability when the base rate is low. Bayes' formula is an accounting identity showing exactly where the effect comes from.

9.3 Independence: learning one event gives no information about another

Definition 9.6. Events A, B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Common pitfall

Disjointness and independence are almost opposites for nontrivial events. If $A \cap B = \emptyset$ and both have positive probability, then $\mathbb{P}(A \cap B) = 0$ but $\mathbb{P}(A)\mathbb{P}(B) > 0$, so they are not independent. Disjoint means one excludes the other; independent means learning one does not change the chance of the other.

Pairwise independence of three events does not imply mutual independence. Mutual independence requires

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i)$$

for every nonempty subset I of indices.

Pairwise but not mutually independent

Flip two fair bits X, Y . Let A be “ $X = 0$,” B be “ $Y = 0$,” and C be “ $X = Y$.” Each event has probability $1/2$, and each pair has intersection probability $1/4$, so the events are pairwise independent. But $A \cap B \subseteq C$, and

$$\mathbb{P}(A \cap B \cap C) = 1/4 \neq 1/8.$$

9.4 Random variables and distributions

Definition 9.7. A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$. Its distribution is the probability measure on $\mathbb{R}$ defined by

$$\mu_X(B) = \mathbb{P}(X \in B).$$

Deep intuition

A random variable is not intrinsically random; it is a deterministic measurement of a random outcome. The same sample point can support many random variables - height, cost, number of successes, waiting time - each extracting a different numerical feature.

For a discrete X taking values x_i with probabilities p_i ,

$$\mathbb{E}[X] = \sum_i x_i p_i.$$

For a random variable with density f , Chapter 9 justifies

$$\mathbb{E}[X] = \int_{\mathbb{R}} xf(x) dx$$

when the integral exists.

9.5 Expectation: linear averaging

Theorem 9.8 (Linearity of expectation). *If X, Y are integrable and $a, b \in \mathbb{R}$, then*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

No independence is required.

Proof. In the discrete case this follows by distributing sums. In the general case it is linearity of the Lebesgue integral, proved in Chapter 9. $\square$

Deep intuition

Expectation is linear because averaging commutes with superposition. This makes expectation much easier than probability of unions or distributions of sums. When faced with a count, try writing it as a sum of indicator variables before attempting casework.

Indicators turn counting into algebra

Let X be the number of fixed points of a uniformly random permutation of $\{1, \dots, n\}$. Write

$$X = I_1 + \dots + I_n,$$

where $I_j = 1$ if the permutation fixes j . Then $\mathbb{E}[I_j] = 1/n$, so

$$\mathbb{E}[X] = \sum_{j=1}^n \frac{1}{n} = 1.$$

No distribution of X was needed.

9.6 Variance and covariance: measuring fluctuation

Definition 9.9. For square-integrable X ,

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

For X, Y ,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

Proposition 9.10.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

If X, Y are independent and square-integrable, then $\text{Cov}(X, Y) = 0$.

Proof. Expand the square after subtracting means and use linearity of expectation. Under independence, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, so covariance vanishes. $\square$

Common pitfall

Zero covariance does not generally imply independence. Covariance detects only linear dependence. For example, if X is symmetric about 0 and nondegenerate and $Y = X^2$, then X and Y are strongly dependent, but $\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ when the odd moments vanish.

9.7 Markov and Chebyshev inequalities

Theorem 9.11 (Markov). *If $X \geq 0$ and $a > 0$, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}.$$

Proof. Pointwise,

$$X \geq a\mathbf{1}_{\{X \geq a\}}.$$

Taking expectations gives $\mathbb{E}X \geq a\mathbb{P}(X \geq a)$. $\square$

Theorem 9.12 (Chebyshev). *If X has finite variance, then for $t > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

Proof. Apply Markov's inequality to the nonnegative variable $(X - \mathbb{E}X)^2$ with threshold t^2 . $\square$

Deep intuition

These inequalities convert moment information into tail information. They are deliberately crude but extremely robust: knowing only an average or a variance already forbids too much probability mass from living very far away.

9.8 Weak law of large numbers

Theorem 9.13 (Weak law, iid finite-variance form). *Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ and variance $\sigma^2 < \infty$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq \varepsilon\right) \rightarrow 0.$$

Proof. Set $\bar{X}_n = n^{-1} \sum_i X_i$. Linearity gives $\mathbb{E}\bar{X}_n = \mu$. Independence gives

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_i \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

Chebyshev therefore yields

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

□

Deep intuition

Averaging does not reduce the mean signal, but independent fluctuations partially cancel. The variance of the average shrinks like $1/n$, so the typical scale of the error shrinks like $1/\sqrt{n}$. The law of large numbers is a stability theorem for repeated independent averaging.

9.9 Probability generating functions

For a nonnegative integer-valued random variable X , define

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k \geq 0} \mathbb{P}(X = k) s^k.$$

When differentiation is justified,

$$G'_X(1) = \mathbb{E}X.$$

If X, Y are independent, then

$$G_{X+Y}(s) = G_X(s)G_Y(s).$$

Connection

The product law is exactly the same convolution-to-multiplication phenomenon as ordinary generating functions in combinatorics and Fourier transforms in analysis. Encodings are powerful when a difficult operation on objects becomes a simple algebraic operation after transformation.

9.10 Conditional expectation in the finite setting

Suppose a finite partition $B_1, \dots, B_k$ represents the information we know. Define

$$\mathbb{E}[X \mid \mathcal{G}](\omega) = \mathbb{E}[X \mid B_j] \quad \text{when } \omega \in B_j,$$

where $\mathcal{G}$ is the sigma-algebra generated by the partition.

Deep intuition

Conditional expectation is the best average prediction compatible with the information currently visible. It is not merely a number; it is a random variable that is constant on outcomes our information cannot distinguish.

The tower property in this finite setting is

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X],$$

which follows from the law of total expectation.

9.11 Markov chains: random dynamics with local memory

A finite-state Markov chain has transition matrix $P = (p_{ij})$ with $p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$ and rows summing to 1. If π_n is the row vector of state probabilities, then

$$\pi_{n+1} = \pi_n P, \quad \pi_n = \pi_0 P^n.$$

A stationary distribution satisfies $\pi P = \pi$.

Connection

This is linear algebra in probabilistic clothing. Long-run behavior is governed by eigenvalues of P , and a stationary distribution is a left eigenvector for eigenvalue 1. Probability and dynamics meet through matrix powers.

9.12 Recall drill

Recall drill

Explain conditioning as renormalizing a restricted universe; independence as unchanged information; random variables as measurements; expectation as linear averaging; indicators as counting devices; variance as mean-square fluctuation; Chebyshev as moment-to-tail conversion; and the weak law as shrinking variance of averages.

9.13 Exercises

Exercise 9.1. Three boxes are chosen with probabilities $1/2, 1/3, 1/6$. Their probabilities of producing a red ball are $1/4, 1/2, 3/4$ respectively. Given that a red ball was observed, compute the posterior probability for each box.

Exercise 9.2. Let $X \sim \text{Bin}(n, p)$. Write X as a sum of indicators and derive $\mathbb{E}X = np$ and $\text{Var}(X) = np(1 - p)$.

Exercise 9.3. If X, Y are independent with finite second moments, prove $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$ from the joint distribution in the discrete case.

Exercise 9.4. A random variable has mean 10 and variance 4. Give the strongest bound Chebyshev alone provides for $\mathbb{P}(|X - 10| \geq 5)$.

Exercise 9.5. Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Compute $\text{Cov}(X, Y)$ and show directly that X, Y are not independent.

Exercise 9.6. For the transition matrix

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix},$$

find the stationary distribution and explain why eigenvalue 1 must always occur for a stochastic matrix.

Measure theory: size, integration, and limits

Riemann integration works beautifully for continuous functions on intervals, but it becomes fragile when limits create discontinuities, when domains are irregular, or when probability requires uncountable sample spaces. Measure theory reorganizes integration around a more stable idea: first decide which sets have a consistent notion of size, then build integrals from those sets upward.

10.1 Why sigma-algebras appear

Definition 10.1. A sigma-algebra $\mathcal{F}$ on a set X is a collection of subsets satisfying

1. $X \in \mathcal{F}$;
2. $A \in \mathcal{F} \Rightarrow X \setminus A \in \mathcal{F}$;
3. $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Its members are called measurable sets.

Countable intersections also belong to $\mathcal{F}$ by De Morgan's law.

Motivation

Why countable operations rather than arbitrary ones? Analysis is built from sequences: limits, countable sums, and approximation procedures. Sigma-algebras are exactly stable enough to survive those operations. Requiring closure under arbitrary unions would often force too many sets to be measurable and destroy the possibility of a countably additive translation-invariant length on all subsets of $\mathbb{R}$.

Generated sigma-algebras

The Borel sigma-algebra $\mathcal{B}(\mathbb{R})$ is the smallest sigma-algebra containing all open sets. Since open intervals with rational endpoints form a countable basis, it is also generated by those intervals. This is an important compression: every Borel set, however complicated, is assembled from a countable family of elementary intervals using countable set operations.

10.2 Measures and countable additivity

Definition 10.2. A measure on $(X, \mathcal{F})$ is a function $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ such that for pairwise disjoint measurable sets E_n ,

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

Deep intuition

Countable additivity is the conservation law of measure: if a set is decomposed into nonoverlapping measurable pieces, its total mass is exactly the sum of the masses of the pieces. Everything else - monotonicity, continuity of measure, integration - grows out of this one principle.

Proposition 10.3 (Basic consequences). *If $A \subseteq B$ are measurable, then $\mu(A) \leq \mu(B)$. If additionally $\mu(A) < \infty$, then*

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

For any measurable A_n ,

$$\mu\left(\bigcup_n A_n\right) \leq \sum_n \mu(A_n).$$

Proof. Since $B = A \sqcup (B \setminus A)$, additivity gives monotonicity and the difference formula. For subadditivity, define disjoint sets

$$B_1 = A_1, \quad B_n = A_n \setminus \bigcup_{j < n} A_j.$$

Then $\bigcup B_n = \bigcup A_n$ and $B_n \subseteq A_n$. Hence

$$\mu\left(\bigcup A_n\right) = \sum \mu(B_n) \leq \sum \mu(A_n).$$

□

10.3 Continuity of measure

Theorem 10.4 (Continuity from below). *If $E_1 \subseteq E_2 \subseteq \dots$, then*

$$\mu\left(\bigcup_n E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Proof. Write

$$E_1 \sqcup (E_2 \setminus E_1) \sqcup (E_3 \setminus E_2) \sqcup \dots = \bigcup_n E_n.$$

Countable additivity shows that $\mu(E_n)$ is the n th partial sum of the series giving the measure of the union. □

Theorem 10.5 (Continuity from above). *If $E_1 \supseteq E_2 \supseteq \cdots$ and $\mu(E_1) < \infty$, then*

$$\mu\left(\bigcap_n E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

Proof. Apply continuity from below to $E_1 \setminus E_n$, which increase to $E_1 \setminus \bigcap_n E_n$. Since $\mu(E_1) < \infty$,

$$\mu(E_1) - \mu(E_n) \rightarrow \mu(E_1) - \mu(\bigcap_n E_n),$$

and cancellation gives the result. $\square$

Common pitfall

The finiteness hypothesis in continuity from above matters. On $\mathbb{R}$ with Lebesgue measure, $E_n = [n, \infty)$ decreases to $\emptyset$, but every $\mu(E_n) = \infty$. The symbolic expression “ $\infty - \infty$ ” is precisely where the proof would fail.

10.4 Lebesgue measure: extending length

The desired measure m on $\mathbb{R}$ should satisfy $m([a, b]) = b - a$, translation invariance, and countable additivity. The construction proceeds through outer measure.

For any $A \subseteq \mathbb{R}$, define

$$m^*(A) = \inf \left\{ \sum_{k=1}^{\infty} |I_k| : A \subseteq \bigcup_{k=1}^{\infty} I_k, I_k \text{ open intervals} \right\}.$$

This is an *outer measure*: it is defined for every subset, but it is only subadditive, not additive on arbitrary sets.

A set E is Caratheodory measurable if for every $A \subseteq \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E).$$

The measurable sets form a sigma-algebra, and m^* restricted to them is countably additive. This produces Lebesgue measure.

Proof architecture

The Caratheodory condition says that cutting any test set A along E creates no hidden loss of outer mass. Subadditivity always gives “ $\leq$ ” in the opposite direction; measurability demands the matching reverse inequality. The remarkable theorem is that sets passing this universal cutting test are closed under complements and countable unions, and outer measure becomes a genuine measure on them.

Theorem 10.6 (Caratheodory construction). *For any outer measure μ^* on X , the collection*

$$\mathcal{M} = \{E \subseteq X : \mu^*(A) = \mu^*(A \cap E) + \mu^*(A \setminus E) \text{ for every } A \subseteq X\}$$

is a sigma-algebra, and $\mu = \mu^|_{\mathcal{M}}$ is a complete measure.*

Proof. The empty set is measurable, and the defining equality is symmetric in E and E^c , so complements preserve measurability. If E, F are measurable and disjoint, apply the condition for E to A , then the condition for F to $A \setminus E$, obtaining

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap F) + \mu^*(A \setminus (E \cup F)),$$

which shows $E \cup F$ is measurable and gives finite additivity on disjoint measurable sets. Inductively this holds for finite disjoint unions.

Now let E_n be pairwise disjoint measurable sets and $E = \bigcup_n E_n$. Put $F_N = \bigcup_{n \leq N} E_n$. Repeated measurability gives for every A

$$\mu^*(A) \geq \sum_{n=1}^N \mu^*(A \cap E_n) + \mu^*(A \setminus E).$$

Letting $N \rightarrow \infty$ and using subadditivity on $A \cap E = \bigcup_n (A \cap E_n)$ yields

$$\mu^*(A) \geq \mu^*(A \cap E) + \mu^*(A \setminus E).$$

The reverse inequality is subadditivity, so E is measurable. Arbitrary countable unions reduce to disjoint ones by replacing E_n with $E_n \setminus \bigcup_{j < n} E_j$. Thus $\mathcal{M}$ is a sigma-algebra.

Taking $A = E$ in the finite-union equality and passing to the union argument gives countable additivity of μ^* on disjoint measurable sets. Finally, if $N \subseteq E \in \mathcal{M}$ with $\mu(E) = 0$, then $\mu^*(N) = 0$ by monotonicity. For every A , subadditivity gives

$$\mu^*(A) \leq \mu^*(A \cap N) + \mu^*(A \setminus N) = 0 + \mu^*(A \setminus N) \leq \mu^*(A),$$

so equality holds and N is measurable. Hence the measure is complete. $\square$

10.5 Measurable functions

Definition 10.7. A function $f : X \rightarrow \mathbb{R}$ is measurable if $f^{-1}((a, \infty)) \in \mathcal{F}$ for every $a \in \mathbb{R}$.

Equivalent tests may use open sets, closed sets, or intervals. Continuous functions between topological spaces equipped with their Borel sigma-algebras are measurable because inverse images of open sets are open.

Proposition 10.8. *If f_n are measurable, then $\sup_n f_n$, $\inf_n f_n$, $\limsup f_n$, and $\liminf f_n$ are measurable where defined.*

Proof. For example,

$$\{x : \sup_n f_n(x) > a\} = \bigcup_n \{x : f_n(x) > a\},$$

which is measurable. Infimum follows by complements or a countable intersection. Limsup and liminf are built from countable infima and suprema:

$$\limsup_n f_n = \inf_N \sup_{n \geq N} f_n, \quad \liminf_n f_n = \sup_N \inf_{n \geq N} f_n.$$

$\square$

Deep intuition

Measurability means the numerical questions asked of f can be translated back into measurable questions about the underlying space. Asking “is $f > a$?” must produce an event whose size is defined.

10.6 Simple functions and the Lebesgue integral

A nonnegative simple function has the form

$$s = \sum_{j=1}^k a_j \mathbf{1}_{E_j}, \quad a_j \geq 0,$$

with measurable E_j . If the E_j are disjoint, define

$$\int s \, d\mu = \sum_{j=1}^k a_j \mu(E_j).$$

For a nonnegative measurable f , define

$$\int f \, d\mu = \sup \left\{ \int s \, d\mu : 0 \leq s \leq f, \, s \text{ simple} \right\}.$$

For general f , write $f = f^+ - f^-$ with $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. If both integrals are finite, f is integrable and

$$\int f = \int f^+ - \int f^-.$$

Motivation

Riemann integration partitions the *domain* into intervals. Lebesgue integration first approximates the *range*: it groups points according to the value of the function. This makes it much better suited to limits in which sets move around while function values remain controlled.

Indicator functions are the atoms of integration

For a measurable set E ,

$$\int \mathbf{1}_E \, d\mu = \mu(E).$$

Thus integration extends measure from sets to functions. A simple function is just a finite weighted stack of measurable sets, and its integral is weighted mass.

10.7 Monotone convergence

Theorem 10.9 (Monotone convergence theorem). *If $0 \leq f_1 \leq f_2 \leq \dots$ are measurable and $f_n(x) \uparrow f(x)$ pointwise, then*

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

Proof. Monotonicity of the integral gives $\int f_n \leq \int f$ and shows the limit $L = \lim_n \int f_n$ exists with $L \leq \int f$.

To prove the reverse inequality, let s be a nonnegative simple function with $s \leq f$, and fix $0 < c < 1$. Define

$$E_n = \{x : f_n(x) \geq cs(x)\}.$$

The sets increase. At every point where $s(x) > 0$, we have $f(x) \geq s(x) > cs(x)$, so eventually $f_n(x) \geq cs(x)$; points where $s = 0$ are irrelevant. Hence E_n increases to a set containing the support of s . Since $f_n \geq cs \mathbf{1}_{E_n}$,

$$\int f_n \geq c \int_{E_n} s.$$

For a simple $s = \sum a_j \mathbf{1}_{A_j}$, continuity from below gives $\int_{E_n} s \rightarrow \int s$. Thus $L \geq c \int s$. Let $c \uparrow 1$ to obtain $L \geq \int s$. Taking the supremum over all simple $s \leq f$ yields $L \geq \int f$. $\square$

Deep intuition

Monotone convergence works because nothing ever has to be undone. Mass only accumulates. The integral can therefore follow the pointwise limit without fear that positive and negative pieces are cancelling or that mass is escaping through oscillation.

10.8 Fatou and dominated convergence

Theorem 10.10 (Fatou's lemma). *If $f_n \geq 0$ are measurable, then*

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

Proof. Let $g_N = \inf_{n \geq N} f_n$. Then $g_N \uparrow \liminf f_n$. By monotone convergence,

$$\int \liminf f_n = \lim_N \int g_N.$$

Since $g_N \leq f_n$ for every $n \geq N$,

$$\int g_N \leq \inf_{n \geq N} \int f_n.$$

Take $N \rightarrow \infty$. $\square$

Theorem 10.11 (Dominated convergence theorem). *If $f_n \rightarrow f$ pointwise almost everywhere and $|f_n| \leq g$ for an integrable g , then f is integrable and*

$$\int |f_n - f| d\mu \rightarrow 0, \quad \int f_n d\mu \rightarrow \int f d\mu.$$

Proof. Pointwise, $|f| \leq g$ almost everywhere, so f is integrable. Apply Fatou to the nonnegative functions $2g - |f_n - f|$:

$$\int 2g \leq \liminf_n \int (2g - |f_n - f|) = 2 \int g - \limsup_n \int |f_n - f|.$$

Thus $\limsup_n \int |f_n - f| \leq 0$, proving L^1 convergence. Finally

$$\left| \int f_n - \int f \right| \leq \int |f_n - f| \rightarrow 0.$$

□

Common pitfall

Pointwise convergence alone does not justify exchanging limit and integral. A classic “moving spike” is

$$f_n(x) = n \mathbf{1}_{(0,1/n)}(x)$$

on $(0,1)$. Then $f_n(x) \rightarrow 0$ for every $x > 0$, but $\int_0^1 f_n = 1$ for all n . The mass concentrates into a shrinking region instead of disappearing.

10.9 Almost everywhere: ignoring null sets coherently

A property holds almost everywhere if it fails only on a set of measure zero.

Deep intuition

Measure theory regards two functions that differ only on a null set as observationally indistinguishable for integration. This is another quotient idea: in L^p spaces, functions are literally identified when they agree almost everywhere.

10.10 L^p spaces and Holder’s inequality

For $1 \leq p < \infty$,

$$L^p = \left\{ f : \int |f|^p < \infty \right\} / \text{a.e. equality}, \quad \|f\|_p = \left(\int |f|^p \right)^{1/p}.$$

If $1 < p, q < \infty$ and $1/p + 1/q = 1$, Holder’s inequality states

$$\int |fg| \leq \|f\|_p \|g\|_q.$$

Theorem 10.12 (Holder). *The inequality above holds for $f \in L^p$ and $g \in L^q$.*

Proof. If either norm is zero, the claim is trivial. Normalize so $\|f\|_p = \|g\|_q = 1$. Young’s inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad (a, b \geq 0)$$

follows from convexity of $t \mapsto t^p/p$ or elementary calculus. Apply it pointwise to $|f|$ and $|g|$ and integrate:

$$\int |fg| \leq \frac{1}{p} \int |f|^p + \frac{1}{q} \int |g|^q = \frac{1}{p} + \frac{1}{q} = 1.$$

Rescaling gives the general result. □

10.11 Modes of convergence

Common notions include:

- pointwise almost-everywhere convergence;
- convergence in measure;
- L^p convergence;
- almost-sure convergence for random variables.

They are not interchangeable. On a finite-measure space, L^p convergence implies convergence in measure by Markov's inequality:

$$\mu(|f_n - f| > \varepsilon) \leq \frac{\|f_n - f\|_p^p}{\varepsilon^p}.$$

Connection

The same phenomenon appeared in uniform versus pointwise convergence. Mathematics has many ways to say “ f_n approaches f ,” each designed to preserve different operations. The right notion of convergence is determined by what theorem you want to pass to the limit.

10.12 Recall drill

Recall drill

Explain why sigma-algebras are countably closed; countable additivity as conservation of mass; continuity from above and why finiteness matters; measurable functions as measurable questions; simple functions as weighted sets; MCT as accumulating mass; dominated convergence as control against escaping spikes; and almost-everywhere equality as a quotient relation.

10.13 Exercises

Exercise 10.1. Prove that if A_n are measurable, then

$$\mu(\liminf A_n) \leq \liminf \mu(A_n),$$

where $\liminf A_n = \bigcup_N \bigcap_{n \geq N} A_n$. Identify this as Fatou's lemma for indicators.

Exercise 10.2. Show that if $f_n \rightarrow f$ uniformly on a finite-measure space and the functions are bounded, then $f_n \rightarrow f$ in L^1 .

Exercise 10.3. On $(0, 1)$, study $f_n(x) = n^2 x(1 - nx)$ for $0 < x < 1/n$ and 0 otherwise. Determine pointwise convergence and the behavior of the integrals.

Exercise 10.4. Prove Young's inequality $ab \leq a^p/p + b^q/q$ for conjugate exponents $p, q > 1$ by convexity.

Exercise 10.5. If $f_n \rightarrow f$ in L^1 , prove $\int f_n \rightarrow \int f$. Give an example showing the converse is false.

Exercise 10.6. Let X_n be random variables with $\mathbb{E}|X_n - X|^2 \rightarrow 0$. Prove $X_n \rightarrow X$ in probability.

Differential equations and dynamical systems: local rules, global motion

A differential equation specifies how a state is allowed to change at each instant. The central conceptual problem is therefore not “how do I integrate this formula?” but “when does a local velocity law determine a unique trajectory, how long does that trajectory exist, and what qualitative features can be inferred without solving explicitly?”

11.1 Vector fields and trajectories

An autonomous ordinary differential equation has the form

$$x'(t) = f(x(t)), \quad x(t) \in \mathbb{R}^n.$$

The function f is a vector field: at each state x , it specifies an instantaneous velocity vector.

Deep intuition

Think of the differential equation as a field of arrows painted on state space. A solution curve is not free to choose its direction; at every point its tangent must align with the arrow painted there. Existence asks whether a curve can follow the arrows. Uniqueness asks whether two curves can pass through the same state and then separate.

11.2 Integral formulation

If x is differentiable and solves

$$x'(t) = f(t, x(t)), \quad x(t_0) = x_0,$$

then the fundamental theorem of calculus gives

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) \, ds.$$

Conversely, a continuous function satisfying this integral equation is differentiable and solves the ODE when f is continuous.

Motivation

The integral equation is more useful for existence proofs because it turns the problem of finding a derivative-constrained curve into a fixed-point problem. Instead of guessing a solution, we search for a function unchanged by the operator

$$(Tx)(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

11.3 Contraction mapping theorem

Theorem 11.1 (Banach fixed-point theorem). *Let (X, d) be a complete metric space and $T : X \rightarrow X$ satisfy*

$$d(Tx, Ty) \leq qd(x, y)$$

for some $0 < q < 1$. Then T has a unique fixed point x^ . For any x_0 , the iterates $x_{n+1} = Tx_n$ converge to x^* .*

Proof. Let $x_{n+1} = Tx_n$. Then

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

For $m > n$, the triangle inequality gives

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} q^j d(x_1, x_0) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

Thus (x_n) is Cauchy and converges by completeness to some x^* . Since a contraction is continuous,

$$Tx^* = T(\lim x_n) = \lim Tx_n = \lim x_{n+1} = x^*.$$

If y^* were another fixed point,

$$d(x^*, y^*) = d(Tx^*, Ty^*) \leq qd(x^*, y^*),$$

forcing the distance to be zero. □

Deep intuition

A contraction forgets initial differences exponentially fast. Repeated application cannot wander: every pair of iterates is pulled together. Fixed-point theorems are powerful because many nonlinear equations can be rewritten as “find an object unchanged by this operator.”

11.4 Picard-Lindelof existence and uniqueness

Theorem 11.2 (Local existence and uniqueness). *Suppose $f(t, x)$ is continuous on a rectangle around (t_0, x_0) and is Lipschitz in x there: for some L ,*

$$\|f(t, x) - f(t, y)\| \leq L \|x - y\|.$$

Then on a sufficiently small interval around t_0 , the initial-value problem

$$x' = f(t, x), \quad x(t_0) = x_0$$

has a unique solution.

Proof. Choose a closed ball B around x_0 inside the rectangle and let M bound $\|f\|$ there. Choose $h > 0$ small enough that Mh is at most the radius of B and $Lh < 1$. Let $X = C([t_0 - h, t_0 + h], B)$ with the sup norm; this is complete. Define

$$(Tx)(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

The choice Mh ensures T maps X into itself. Moreover

$$\|Tx - Ty\|_\infty \leq hL \|x - y\|_\infty.$$

Thus T is a contraction. Banach's theorem gives a unique fixed point, which is exactly a solution of the integral equation and hence the ODE. $\square$

Common pitfall

Continuity of f alone gives existence under broader theorems but not uniqueness. The scalar equation

$$x' = 2\sqrt{|x|}, \quad x(0) = 0$$

has the zero solution, but also solutions that remain at zero for an arbitrary waiting time and then depart. The failure occurs because $f(x) = 2\sqrt{|x|}$ is not Lipschitz near 0.

11.5 Gronwall's inequality and stability

Theorem 11.3 (Gronwall, simple form). *If a nonnegative continuous function u satisfies*

$$u(t) \leq A + B \int_0^t u(s) ds$$

for $t \in [0, T]$, with $A, B \geq 0$, then

$$u(t) \leq Ae^{Bt}.$$

Proof. Let

$$v(t) = A + B \int_0^t u(s) ds.$$

Then $u \leq v$ and $v'(t) = Bu(t) \leq Bv(t)$. Hence

$$\frac{d}{dt}(e^{-Bt}v(t)) = e^{-Bt}(v' - Bv) \leq 0.$$

So $e^{-Bt}v(t) \leq v(0) = A$, giving $u(t) \leq v(t) \leq Ae^{Bt}$. $\square$

Connection

Gronwall is the continuous analogue of repeatedly multiplying an error by a growth factor. It is the workhorse behind uniqueness, continuous dependence on initial conditions, numerical-error estimates, and stability theory.

11.6 Linear systems and the matrix exponential

Consider

$$x'(t) = Ax(t), \quad x(0) = x_0.$$

Define

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}.$$

The series converges in any matrix norm, and termwise differentiation gives

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A.$$

Thus

$$x(t) = e^{tA} x_0.$$

Deep intuition

The scalar equation $x' = \lambda x$ has solution $e^{\lambda t} x_0$. The matrix exponential is not a formal imitation; it is the unique operator-valued function whose derivative is left multiplication by A and whose initial value is the identity. Eigenvectors reduce it back to the scalar story.

If $A = PDP^{-1}$, then

$$e^{tA} = P e^{tD} P^{-1}.$$

For a Jordan block $A = \lambda I + N$ with $N^m = 0$,

$$e^{tA} = e^{\lambda t} \sum_{k=0}^{m-1} \frac{t^k N^k}{k!}.$$

This explains why polynomial factors multiply exponentials when diagonalization fails.

11.7 Phase portraits from eigenvalues

For $x' = Ax$ in $\mathbb{R}^2$:

- two negative real eigenvalues give an attracting node;
- two positive real eigenvalues give a repelling node;
- opposite signs give a saddle;
- complex conjugate eigenvalues $\alpha \pm i\beta$ give spiraling when $\alpha \neq 0$ and rotation-like centers in special conservative cases when $\alpha = 0$.

A saddle is visible from eigendirections

For

$$A = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}, \quad x(t) = \begin{pmatrix} c_1 e^{2t} \\ c_2 e^{-t} \end{pmatrix}.$$

Along the y -axis trajectories approach the origin as $t \rightarrow \infty$; any nonzero x -component grows exponentially. The origin is therefore stable in one direction and unstable in another: a saddle.

11.8 Nonlinear systems and linearization

Let $x' = f(x)$ and suppose $f(a) = 0$, so a is an equilibrium. Write $x = a + h$. Then

$$h' = Df(a)h + o(\|h\|).$$

The matrix $Df(a)$ is the linearization.

Deep intuition

Linearization is the dynamical version of the derivative. Near an equilibrium, the nonlinear flow is governed to first order by a linear system. Hyperbolic equilibria - those whose linearization has no eigenvalues with zero real part - are robust because exponential contraction or expansion dominates smaller nonlinear terms.

A full Hartman-Grobman theorem is beyond a reboot proof, but the principle can already be seen in one dimension.

Proposition 11.4 (One-dimensional stability test). *Let $x' = f(x)$ with $f \in C^1$, $f(a) = 0$. If $f'(a) < 0$, then a is locally asymptotically stable. If $f'(a) > 0$, it is unstable.*

Proof. If $f'(a) < 0$, continuity gives a neighborhood in which $(x - a)f(x) < 0$ for $x \neq a$: to the right of a , velocity is negative; to the left, positive. Thus trajectories move toward a . More quantitatively, shrink the neighborhood so $f'(x) \leq -c < 0$. By the mean value theorem, $f(x) = f'(\xi)(x - a)$, hence

$$\frac{d}{dt} \frac{1}{2} (x - a)^2 = (x - a)f(x) \leq -c(x - a)^2.$$

Gronwall gives exponential decay. If $f'(a) > 0$, the sign reverses and nearby trajectories move away on either side. $\square$

11.9 Conserved quantities and Lyapunov functions

A function $H(x)$ is conserved along trajectories if

$$\frac{d}{dt} H(x(t)) = \nabla H(x) \cdot f(x) = 0.$$

A Lyapunov function V instead decreases:

$$\nabla V(x) \cdot f(x) \leq 0.$$

Energy of the harmonic oscillator

For

$$x' = v, \quad v' = -x,$$

let $H(x, v) = \frac{1}{2}(x^2 + v^2)$. Then

$$\frac{d}{dt}H = xx' + vv' = xv + v(-x) = 0.$$

Trajectories remain on circles $x^2 + v^2 = \text{constant}$. The conservation law reveals the phase portrait without explicitly solving the ODE.

Motivation

An invariant can outperform a formula. Exact solutions may be inaccessible, but a conserved energy, monotone entropy, or Lyapunov function can constrain every trajectory at once. This is the continuous-time analogue of invariants and monovariants in discrete mathematics.

11.10 Separation and exact solvability as structural special cases

For $x' = g(t)h(x)$ with $h(x) \neq 0$, formally

$$\frac{dx}{h(x)} = g(t)dt.$$

The rigorous justification is to choose an antiderivative $H'(x) = 1/h(x)$; then the chain rule gives

$$\frac{d}{dt}H(x(t)) = g(t).$$

Thus

$$H(x(t)) = H(x_0) + \int_{t_0}^t g(s) ds.$$

Common pitfall

Dividing by $h(x)$ can discard equilibrium solutions where $h(x) = 0$. Always identify zeros of h before dividing. Algebraic manipulations of differential equations can change the solution set if singular cases are silently removed.

11.11 Recall drill**Recall drill**

Explain a vector field as a local velocity rule; why the integral equation is suited to fixed points; Lipschitz continuity as a uniqueness condition; Gronwall as error-growth control; the matrix exponential as linear flow; eigenvalues as phase-portrait data; linearization as

first-order dynamics; and conserved quantities as invariants.

11.12 Exercises

Exercise 11.1. Use Picard iteration beginning with $x_0(t) = 0$ to generate the first four iterates for $x' = 1 + x$, $x(0) = 0$. Recognize the limiting series.

Exercise 11.2. Solve $x' = Ax$ for $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ and explain the factor of t using Jordan form.

Exercise 11.3. Classify the equilibrium at the origin for each diagonal system $x' = \lambda x$, $y' = \mu y$ according to the signs of λ, μ .

Exercise 11.4. For the logistic equation $x' = rx(1 - x/K)$ with $r, K > 0$, find equilibria and determine their local stability from $f'(x)$.

Exercise 11.5. Show that $H(x, v) = \frac{1}{2}v^2 + U(x)$ is conserved for the system $x' = v$, $v' = -U'(x)$. Explain how level curves of H constrain motion.

Exercise 11.6. Use Gronwall to prove uniqueness directly for solutions of $x' = f(t, x)$ under a Lipschitz condition: estimate the distance between two solutions with the same initial value.

Synthesis: recurring ideas across the mathematical landscape

The purpose of a reboot is not to leave behind ten separate refresher chapters. The real payoff is seeing the same structural moves reappear under different names.

12.1 Local linearization

Linear algebra is not one subject among many; it is the universal first-order language.

- In multivariable calculus, $DF(a)$ is the local linear model of F .
- In constrained optimization, tangent spaces are kernels of differentials.
- In ODEs, $Df(a)$ controls dynamics near an equilibrium.
- In probability, covariance matrices describe second-order linear fluctuation.
- In geometry, tangent spaces replace curved objects by vector spaces at small scale.

Deep intuition

Nonlinear mathematics often proceeds by a two-stage strategy: first understand the linear shadow, then control the error between the shadow and the real object. The derivative, Hessian, tangent space, Jacobian, and linearization are all versions of this strategy at different orders.

12.2 Quotients as deliberate loss of information

The quotient pattern appears repeatedly:

objects $\longrightarrow$ declare some differences invisible $\longrightarrow$ new object.

Examples include V/W , congruence classes $\mathbb{Z}/n\mathbb{Z}$, topological quotient spaces, equivalence classes of loops in π_1 , random variables modulo almost-everywhere equality in L^p , and connected components.

Connection

Whenever a quotient appears, ask three questions: What equivalence relation is being imposed? Why is that relation compatible with the structure we want to keep? Which information survives the quotient? These questions often reveal the theorem before any calculation begins.

12.3 Compactness as controlled finiteness

Compactness is infinite behavior with a finite-control principle hidden inside it. Open covers reduce to finite subcovers; bounded sequences in Euclidean space have convergent subsequences; continuous functions achieve extrema; local continuity becomes uniform continuity.

In finite combinatorics, finiteness itself provides the stopping mechanism. In topology and analysis, compactness is often the replacement for literal finiteness.

Deep intuition

Compactness is not merely “closed and bounded.” That equivalence is a special theorem in Euclidean space. The reusable idea is: an infinite object is compact when local information cannot escape every finite attempt at control.

12.4 Invariants, conserved quantities, and obstruction theory

The checkerboard coloring obstruction, graph parity, determinant, trace, homotopy class, winding number, conserved energy, and total probability all exemplify invariants.

A good invariant compresses a complicated object into a simpler quantity that allowed operations cannot change. If two configurations have different invariant values, no sequence of allowed operations can connect them.

12.5 Duality and measurement

The dual-space viewpoint says an object can be understood by how every linear measurement responds to it. Measure theory similarly understands sets through size and functions through integrals. Probability distributions are determined by families of expectations or transforms under suitable hypotheses.

Deep intuition

There are two complementary ways to understand an object: describe its internal coordinates, or probe it from the outside with measurements. Duality formalizes the second perspective. In advanced mathematics, the space of probes can be easier to organize than the original objects.

12.6 Encoding difficult operations as easy ones

Several transforms share one design principle:

Encoding	Difficult operation	Easier operation after encoding
Matrix representation	composition of linear maps	matrix multiplication
Generating function	recurrence / convolution	algebra / multiplication
Probability generating function	sum of independent integer variables	product
Eigenbasis	repeated linear transformation	scalar powers
Fourier/Laplace viewpoint	differentiation / convolution	multiplication
Quotient	irrelevant distinctions	literal equality of classes

The recurring question is: *Can I represent the objects in a language where the operation I care about becomes simpler?*

12.7 Local-to-global reasoning

Topology and geometry repeatedly ask when compatible local data glue globally. Compactness reduces infinitely many local choices to finitely many. Connectedness prevents local branches from separating. Differential equations build a global trajectory from local velocity rules. Hall's theorem assembles local neighborhood capacity into a global matching.

Motivation

Many deep theorems have the shape “a family of local constraints is globally consistent if and only if the right compatibility condition holds.” Recognizing this shape prepares the ground for manifolds, sheaves, differential geometry, algebraic topology, and algebraic geometry.

12.8 Different notions of convergence protect different operations

Pointwise convergence protects very little. Uniform convergence controls all points simultaneously. L^p convergence controls average magnitude. Convergence in probability controls the probability of large errors. Almost-sure convergence controls individual outcomes outside a null set.

Deep intuition

There is no universally “correct” notion of convergence. A convergence notion is a contract: it gives enough control to pass specific structures to the limit. Before choosing a topology or norm, ask what operations need to remain stable.

12.9 A dependency roadmap for further study

After this reboot, several natural routes open.

- **Functional analysis:** linear algebra + topology + measure theory $\rightarrow$ Banach and Hilbert spaces, bounded operators, spectral theory.
- **Differential geometry:** linear algebra + multivariable calculus + topology $\rightarrow$ manifolds, tangent bundles, differential forms.
- **Algebraic topology:** point-set topology + groups $\rightarrow$ fundamental groups, homology, cohomology.
- **Advanced probability:** measure theory + linear algebra $\rightarrow$ conditional expectation, martingales, stochastic processes.
- **Combinatorics and algorithms:** discrete mathematics + graph theory + linear algebra + probability $\rightarrow$ probabilistic method, spectral graph theory, randomized algorithms.
- **Partial differential equations:** multivariable calculus + ODEs + measure theory + functional analysis.
- **Number theory:** algebra + analysis + topology $\rightarrow$ algebraic, analytic, and arithmetic geometry viewpoints.

12.10 Final mastery questions

A student ready to leave this reboot should be able to answer these without notes.

1. Why is a matrix secondary to a linear map? What data are basis-dependent and what are intrinsic?
2. Why is the derivative a linear map rather than a list of partial derivatives?
3. Why does topology define continuity with inverse images of open sets?
4. What is compactness actually buying in proofs?
5. How do bijections, invariants, and extremal choices differ as proof strategies?
6. Why do trees sit exactly at minimal connectivity?
7. Why is modular arithmetic naturally quotient arithmetic?
8. Why does independence cause variances to add?
9. Why is dominated convergence stronger than pointwise convergence?
10. Why does a Lipschitz condition lead to uniqueness for an ODE?
11. Give three mathematically different examples of a quotient and identify what information is forgotten.
12. Give three examples where an invariant turns a hard evolution problem into a static obstruction.

Proof architecture

For any theorem you retain from this book, store a three-part skeleton: *input hypotheses*, *decisive mechanism*, *output*. Examples: rank-nullity = extend a kernel basis; compact image = pull back an open cover; Hall = induct on a tight set or delete one matched pair; weak law = variance of average plus Chebyshev; Picard-Lindelof = integral operator plus contraction.

Complete solutions

These solutions are intentionally written as proof reconstructions rather than answer keys. Each begins from the structural move that makes the problem work.

A.1 Solutions to the baseline diagnostic

D1: linear maps

Rank-nullity gives

$$4 = \dim \ker T + \dim \operatorname{im} T = 2 + \dim \operatorname{im} T,$$

so $\dim \operatorname{im} T = 2$. Since the codomain has dimension 3, the image cannot be all of $\mathbb{R}^3$, so T is not surjective. Since the kernel is nonzero, T is not injective. The basis-free theorem is: for a linear map $T : V \rightarrow W$ with V finite-dimensional,

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

D2: eigenvalues and geometry

For

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \quad \det(A - \lambda I) = (2 - \lambda)^2,$$

so the only eigenvalue is 2 with algebraic multiplicity 2. The eigenspace is

$$\ker(A - 2I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \operatorname{span}\{(1, 0)\},$$

which has dimension 1. Hence there is no eigenbasis and A is not diagonalizable. Geometrically, $A = 2I + N$ with $N(x, y) = (y, 0)$: besides doubling, the map shears the vertical component into the horizontal direction.

D3: orthogonal projection

The plane has normal vector $n = (1, 1, 1)$. Write $v = (1, 2, 3)$. The component of v along n is

$$\frac{v \cdot n}{n \cdot n} n = \frac{6}{3}(1, 1, 1) = (2, 2, 2).$$

Subtracting gives the component orthogonal to n , hence lying in the plane:

$$P(v) = v - \frac{v \cdot n}{n \cdot n} n = (-1, 0, 1).$$

The formula is derived by requiring $v - cn$ to lie in the plane: $(v - cn) \cdot n = 0$, so $c = (v \cdot n)/(n \cdot n)$.

D4: derivative as a linear map

For $F(x, y) = (x^2 - y, xy)$,

$$DF(x, y) = \begin{pmatrix} 2x & -1 \\ y & x \end{pmatrix}, \quad DF(1, 2) = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}.$$

Since $F(1, 2) = (-1, 2)$,

$$F(1 + h, 2 + k) = (-1, 2) + (2h - k, 2h + k) + o(\sqrt{h^2 + k^2}).$$

Thus the derivative predicts the first-order displacement of the output.

D5: compactness and uniform continuity

Fix $\varepsilon > 0$. For each $x \in [0, 1]$, continuity gives $r_x > 0$ such that $|y - x| < 2r_x$ implies $|f(y) - f(x)| < \varepsilon/2$. The intervals $(x - r_x, x + r_x)$ cover $[0, 1]$. Compactness produces finitely many, with radii $r_1, \dots, r_N$. Let $\delta = \min r_j > 0$. If $|u - v| < \delta$, choose a selected center x_j with $|u - x_j| < r_j$. Then $|v - x_j| < 2r_j$, so both $f(u)$ and $f(v)$ lie within $\varepsilon/2$ of $f(x_j)$. Hence $|f(u) - f(v)| < \varepsilon$. Compactness enters exactly when the infinitely many local radii are reduced to finitely many so their minimum stays positive.

D6: topological continuity

Topological continuity is

$$U \subseteq Y \text{ open} \implies f^{-1}(U) \subseteq X \text{ open}.$$

For metric spaces, the equivalence with ε - δ continuity is proved in Chapter 5: one direction applies ε - δ continuity to a ball contained in U ; the other uses openness of $f^{-1}(B_\varepsilon(f(x)))$ to find a δ -ball around x contained in that inverse image.

D7: counting surjections

There are 3^5 functions from a five-element set to a three-element set. Let A_i be the functions missing target value i . Then $|A_i| = 2^5$, $|A_i \cap A_j| = 1$, and the triple intersection is empty. Inclusion-exclusion gives

$$3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150.$$

D8: trees

Take a longest simple path in a finite tree with at least two vertices. If an endpoint had a neighbor outside the path, the path could be extended; if it had another neighbor on the path, a cycle would result. Thus each endpoint has degree 1, giving at least two leaves.

If G is a tree, repeatedly remove a leaf. Induction gives $|E| = |V| - 1$. Conversely, if G is connected with $|V| - 1$ edges and had a cycle, deleting a cycle edge would preserve connectivity and give a connected graph with fewer than $|V| - 1$ edges. But every connected n -vertex graph contains a spanning tree and therefore has at least $n - 1$ edges. Contradiction.

D9: congruences

The Euclidean algorithm gives

$$1 = 2 \cdot 43 - 5 \cdot 17,$$

so $17^{-1} \equiv -5 \equiv 38 \pmod{43}$. Hence

$$x \equiv 5 \cdot 38 \equiv 190 \equiv 18 \pmod{43}.$$

In general, $ax \equiv b \pmod{n}$ is solvable exactly when $\gcd(a, n) \mid b$.

D10: variance of a sum

Let $\mu_X = \mathbb{E}X$ and $\mu_Y = \mathbb{E}Y$. Expanding,

$$\text{Var}(X + Y) = \mathbb{E}[(X - \mu_X + Y - \mu_Y)^2] = \text{Var } X + \text{Var } Y + 2 \text{Cov}(X, Y).$$

If X, Y are independent, then $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$, so $\text{Cov}(X, Y) = 0$. Without independence, the covariance term remains.

D11: continuity from above

Take $E_n = [n, \infty)$ in $\mathbb{R}$ with Lebesgue measure. Then $E_n \downarrow \emptyset$, but every $m(E_n) = \infty$. Thus the formal equality would ask us to interpret a limit of infinite values as 0. The correct theorem is: if $E_n \downarrow E$ and $\mu(E_1) < \infty$, then $\mu(E_n) \downarrow \mu(E)$.

D12: linear dynamics

For $A = \text{diag}(2, -1)$,

$$x_1(t) = c_1 e^{2t}, \quad x_2(t) = c_2 e^{-t}.$$

The first eigendirection expands and the second contracts. The origin is a saddle: the x_2 -axis is the stable direction, while any nonzero x_1 component is exponentially unstable.

A.2 Solutions to Chapter 2: linear algebra I

Exercise 2.1

For differentiation $T : P_3 \rightarrow P_2$, the kernel consists of constant polynomials, so $\ker T = \text{span}\{1\}$. Every quadratic $a + bx + cx^2$ is the derivative of $ax + (b/2)x^2 + (c/3)x^3$, so $\text{im } T = P_2$. Thus rank-nullity is $4 = 1 + 3$. The induced map

$$P_3 / \text{span}\{1\} \rightarrow P_2, \quad [p] \mapsto p',$$

is well-defined and is an isomorphism by the first isomorphism theorem.

Exercise 2.2

If $Ax = b$ has one solution x_0 , then $Ax = b$ iff

$$A(x - x_0) = 0,$$

so every solution has form $x = x_0 + k$ with $k \in \ker A$. Conversely every such vector solves the equation. Hence the solution set is $x_0 + \ker A$. If $b \neq 0$, then 0 is not a solution, so the set cannot be a vector subspace.

Exercise 2.3

Define $T : U \oplus W \rightarrow U + W$ by $T(u, w) = u + w$. It is surjective. Its kernel consists of pairs (u, w) with $u = -w$, so $u \in U \cap W$ and

$$\ker T = \{(z, -z) : z \in U \cap W\} \cong U \cap W.$$

Rank-nullity gives

$$\dim U + \dim W = \dim(U \cap W) + \dim(U + W),$$

which rearranges to the desired formula.

Exercise 2.4

If $v_1, \dots, v_k$ are independent and $\sum a_i T(v_i) = 0$, then

$$T\left(\sum a_i v_i\right) = 0.$$

Injectivity gives $\sum a_i v_i = 0$, so all $a_i = 0$. Thus images of independent sets remain independent.

Dually, if $w_1, \dots, w_k$ span W and $T : V \rightarrow W$ is surjective, any $y \in W$ is $T(x)$ for some $x = \sum a_i w_i$, hence $y = \sum a_i T(w_i)$. Thus the images span W .

Exercise 2.5

A linear functional on $\mathbb{R}^3$ has form $\varphi(x, y, z) = ax + by + cz$. Vanishing on all vectors with $x + y + z = 0$ forces (a, b, c) to be perpendicular to that plane, hence proportional to $(1, 1, 1)$. Therefore

$$W^\circ = \text{span}\{(x, y, z) \mapsto x + y + z\}.$$

It measures signed displacement in the normal direction to the plane.

A.3 Solutions to Chapter 3: linear algebra II**Exercise 3.1**

If $A = (a_{ij})$ and $B = (b_{ij})$, then

$$\text{tr}(AB) = \sum_i (AB)_{ii} = \sum_{i,j} a_{ij} b_{ji} = \sum_{j,i} b_{ji} a_{ij} = \text{tr}(BA).$$

Hence if $B = P^{-1}AP$,

$$\text{tr}(B) = \text{tr}(P^{-1}AP) = \text{tr}(APP^{-1}) = \text{tr} A.$$

Exercise 3.2

For

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix},$$

$\chi_A(\lambda) = \lambda(\lambda - 5)$, so the eigenvalues are 0 and 5. An eigenvector for 5 is $(2, 1)$; for 0 one is $(1, -2)$. After normalization,

$$e_1 = \frac{1}{\sqrt{5}}(2, 1), \quad e_2 = \frac{1}{\sqrt{5}}(1, -2).$$

Because one eigenvalue is 0, A is positive semidefinite, not positive definite. The level set $x^T A x = 1$ is not an ellipse: it is the pair of parallel lines determined by $5y_1^2 = 1$ in eigen-coordinates, reflecting degeneracy in the zero-eigenvalue direction.

Exercise 3.3

Take $v_1 = (1, 1, 0)$ and $v_2 = (0, 1, 1)$. Let

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0).$$

Subtract the projection of v_2 onto e_1 :

$$u_2 = v_2 - \frac{1}{\sqrt{2}}e_1 = (-1/2, 1/2, 1), \quad \|u_2\| = \sqrt{3/2}.$$

Thus

$$e_2 = \frac{1}{\sqrt{6}}(-1, 1, 2).$$

The projection matrix is

$$P = e_1 e_1^T + e_2 e_2^T = \frac{1}{3} \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}.$$

Exercise 3.4

If $Tu = \lambda u$ and $Tv = \mu v$ with $T = T^*$, then

$$\lambda \langle u, v \rangle = \langle Tu, v \rangle = \langle u, Tv \rangle = \mu \langle u, v \rangle.$$

If $\lambda \neq \mu$, then $\langle u, v \rangle = 0$.

Exercise 3.5

Write $A = 2I + N$ with

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

Since $2I$ and N commute,

$$A^n = (2I + N)^n = 2^n I + n2^{n-1} N = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}.$$

The factor n records accumulated shear: the nilpotent component is repeatedly carried along by the dominant scaling.

A.4 Solutions to Chapter 4: multivariable calculus

Exercise 4.1

For a linear map T ,

$$T(a + h) - T(a) - T(h) = 0.$$

Thus the remainder is identically zero, certainly $o(\|h\|)$, so $DT(a) = T$ at every point.

Exercise 4.2

For $f(x, y) = x^2 + 4xy + 5y^2$,

$$H = \begin{pmatrix} 2 & 4 \\ 4 & 10 \end{pmatrix}.$$

The eigenvalues solve $\lambda^2 - 12\lambda + 4 = 0$, so $\lambda = 6 \pm 4\sqrt{2}$, both positive. Hence the Hessian is positive definite and the origin is a strict local minimum. In orthonormal eigen-coordinates, f is a positive weighted sum of two squares, so level sets are ellipses.

Exercise 4.3

For $f(x, y) = xy$ under $g = x^2 + y^2 = 1$,

$$(y, x) = \lambda(2x, 2y).$$

Thus $y = 2\lambda x$ and $x = 2\lambda y$. Nonzero solutions give $4\lambda^2 = 1$ and therefore $y = \pm x$. On the circle, the four candidates are $(\pm 1/\sqrt{2}, \pm 1/\sqrt{2})$. The maximum is $1/2$ when x, y have the same sign; the minimum is $-1/2$ when signs differ.

Alternatively, set $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$. Then $x^2 + y^2 = u^2 + v^2$ and

$$xy = \frac{u^2 - v^2}{2},$$

whose extrema on $u^2 + v^2 = 1$ are visibly $\pm 1/2$.

Exercise 4.4

For $F(x, y) = (e^x \cos y, e^x \sin y)$,

$$DF = e^x \begin{pmatrix} \cos y & -\sin y \\ \sin y & \cos y \end{pmatrix},$$

so $\det DF = e^{2x} > 0$ everywhere. The inverse function theorem gives local invertibility at every point. But $F(x, y + 2\pi) = F(x, y)$, so it is not globally injective.

Exercise 4.5

Let $F(x, y) = x^2 + y^2 - 1$. At $(1, 0)$, $\partial F/\partial x = 2 \neq 0$, so the implicit function theorem solves $x = g(y)$ locally. Differentiate

$$g(y)^2 + y^2 = 1.$$

Then $2gg' + 2y = 0$, so $g'(0) = 0$. Differentiating again,

$$2(g')^2 + 2gg'' + 2 = 0.$$

At 0, $g = 1$ and $g' = 0$, giving $g''(0) = -1$.

A.5 Solutions to Chapter 5: topology**Exercise 5.1**

Every usual open interval (a, b) is a union of half-open intervals $[x, y)$ contained in it, so every usual open set is open in the lower-limit topology. The set $[0, 1)$ itself is basic open in the lower-limit topology but not open in the usual topology. Thus the lower-limit topology is strictly finer.

Exercise 5.2

If A is closed and $x_n \in A$ with $x_n \rightarrow x$, then $x \notin A$ would place x in the open set $X \setminus A$; eventually x_n would lie there, contradiction. Conversely, if A is not closed, choose $x \in \text{cl } A \setminus A$. For each n , the ball $B_{1/n}(x)$ meets A , so choose $x_n \in A \cap B_{1/n}(x)$. Then $x_n \rightarrow x \notin A$.

Exercise 5.3

Let $f : X \rightarrow Y$ be a continuous bijection with X compact and Y Hausdorff. If $F \subseteq X$ is closed, then F is compact. Its image $f(F)$ is compact, hence closed in the Hausdorff space Y . Thus f is a closed bijection, which implies f^{-1} is continuous. Hence f is a homeomorphism.

Exercise 5.4

Fix $(x_0, y_0) \in X \times Y$. For each $x \in X$, the slice $\{x\} \times Y$ is connected as a continuous image of Y , and it meets the connected set $X \times \{y_0\}$ at (x, y_0) . The union of a connected set with any family of connected sets all meeting it is connected. Their union is $X \times Y$.

Exercise 5.5

Let $Q = [0, 1]/(0 \sim 1)$ and let $\bar{q} : Q \rightarrow S^1$ be induced by $q(t) = e^{2\pi it}$. It is a continuous bijection by the universal property of the quotient topology. The space Q is compact because it is the continuous image of compact $[0, 1]$, and S^1 is Hausdorff. By Exercise 5.3, $\bar{q}$ is a homeomorphism.

Exercise 5.6

For $x, y \in \mathbb{R}^n$, the path $\gamma(t) = (1 - t)x + ty$ lies in $\mathbb{R}^n$, so $\mathbb{R}^n$ is path-connected. The homotopy $H(x, t) = (1 - t)x$ contracts it to 0. If a space is path-connected, then any two points lie in the connected image of $[0, 1]$, and standard union reasoning shows the whole space is connected. Therefore a disconnected space cannot be path-connected.

A.6 Solutions to Chapter 6: discrete mathematics and combinatorics

Exercise 6.1

Partition an $(m + n)$ -element set into blocks A of size m and B of size n . Count r -element subsets. Directly there are $\binom{m+n}{r}$. Alternatively choose k elements from A and $r - k$ from B , then sum over k :

$$\binom{m+n}{r} = \sum_k \binom{m}{k} \binom{n}{r-k}.$$

Exercise 6.2

Let A_2, A_3, A_5 be the multiples of 2, 3, 5 up to 1000. Inclusion-exclusion gives

$$500 + 333 + 200 - 166 - 100 - 66 + 33 = 734.$$

Here the pairwise intersections count multiples of 6, 10, 15, and the triple intersection multiples of 30.

Exercise 6.3

For each term a_i , let p_i be the length of the longest increasing subsequence ending at a_i , and q_i the length of the longest decreasing subsequence ending at a_i . If no increasing or decreasing subsequence has length $n+1$, then each pair (p_i, q_i) lies in $\{1, \dots, n\}^2$, giving only n^2 possible pairs. With $n^2 + 1$ terms, two indices $i < j$ have the same pair. If $a_i < a_j$, then $p_j \geq p_i + 1$; if $a_i > a_j$, then $q_j \geq q_i + 1$. Either contradicts equality of pairs.

Exercise 6.4

The characteristic equation is

$$r^2 - 5r + 6 = (r - 2)(r - 3) = 0,$$

so $a_n = A2^n + B3^n$. From $a_0 = 2$ and $a_1 = 5$,

$$A + B = 2, \quad 2A + 3B = 5,$$

so $A = B = 1$. Therefore $a_n = 2^n + 3^n$. In state form,

$$\begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix} = \begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_{n-1} \\ a_{n-2} \end{pmatrix}.$$

Exercise 6.5

Let a_n count binary strings of length n with no consecutive ones. A valid string ends in 0, after which any valid length- $n-1$ prefix is possible, or ends in 10, after which any valid length- $n-2$ prefix is possible. Hence

$$a_n = a_{n-1} + a_{n-2}, \quad a_0 = 1, \quad a_1 = 2.$$

For $A(x) = \sum_{n \geq 0} a_n x^n$,

$$A(x) - 1 - 2x = x(A(x) - 1) + x^2 A(x),$$

so

$$A(x) = \frac{1+x}{1-x-x^2}.$$

Exercise 6.6

Starting from any element in a nonempty finite poset and moving strictly downward whenever possible must terminate, giving a minimal element. Moving strictly upward gives a maximal element. For example, in the divisibility poset on $\{2, 3, 6, 10, 15\}$, the minimal elements are 2, 3 and the maximal elements are 6, 10, 15. There is no minimum because neither 2 nor 3 divides the other, and no maximum because no listed element is divisible by all the others.

A.7 Solutions to Chapter 7: graph theory**Exercise 7.1**

Take a longest simple path $v_0, v_1, \dots, v_k$ in the tree. Since the tree has at least two vertices, $k \geq 1$. If v_0 had a neighbor not equal to v_1 , that neighbor cannot already appear later on the path, because then the edge to it together with the path segment would create a cycle. Thus it would extend the path, contradicting maximality. Therefore $\deg(v_0) = 1$. The same argument applies to v_k .

Exercise 7.2

Let T be a tree and add an edge uv not already present. Since T is connected, there is a unique path from u to v . Together with the new edge, this path forms a cycle. Any cycle containing the new edge must use a path from u to v in the original tree, and that path is unique. Hence exactly one cycle is created.

Exercise 7.3

If $K_{3,3}$ had a planar embedding, every face boundary would have length at least 4 because the graph is bipartite and therefore has no odd cycles. Counting edge-sides gives $4F \leq 2E$, so $F \leq E/2 = 9/2$. Euler's formula would give

$$2 = V - E + F \leq 6 - 9 + \frac{9}{2} = \frac{3}{2},$$

a contradiction. Thus $K_{3,3}$ is nonplanar.

Exercise 7.4

Partition V into $A \sqcup B$ so that the number of crossing edges is maximal. Consider a vertex $v \in A$. If more than half of its incident edges stayed inside A , moving v to B would turn those internal edges into crossing edges and turn fewer crossing edges into internal ones, increasing the total. Contradiction. Thus at least half of the incident edges of each vertex cross. Summing crossing degrees,

$$2E_{\text{cross}} = \sum_v \deg_{\text{cross}}(v) \geq \frac{1}{2} \sum_v \deg(v) = |E|.$$

Hence $E_{\text{cross}} \geq |E|/2$, and the crossing edges form a bipartite subgraph.

Exercise 7.5

Let the bipartition be L, R and assume every vertex has degree k . For $S \subseteq L$, count edges from S to $N(S)$. There are exactly $k|S|$ such incidences from the S side. Each vertex of $N(S)$ is incident to at most k of them, so

$$k|S| \leq k|N(S)|.$$

Thus $|N(S)| \geq |S|$. Hall's theorem gives a matching saturating L . Since $k|L| = |E| = k|R|$, the two parts have equal size, so the matching is perfect.

Exercise 7.6

If the graph has an Euler trail from one odd vertex u to the other v , then every edge lies in that single trail, so certainly in a trail from u to v . Conversely, the extra condition “each edge lies in some u - v trail” by itself does not guarantee that one trail uses every edge; different edges may require different trails. What does guarantee one global trail is the Euler criterion: connectedness together with exactly two odd vertices. Under the hypothesis of the exercise, that criterion already applies, so an Euler trail exists. The point is that an edgewise family of trails is logically weaker than one universal trail; the parity theorem supplies the missing global assembly.

A.8 Solutions to Chapter 8: number theory**Exercise 8.1**

The Euclidean algorithm gives

$$252 = 198 + 54, \quad 198 = 3 \cdot 54 + 36, \quad 54 = 36 + 18, \quad 36 = 2 \cdot 18.$$

Thus $\gcd(252, 198) = 18$. Back-substitution gives

$$18 = 54 - 36 = 54 - (198 - 3 \cdot 54) = 4 \cdot 54 - 198 = 4(252 - 198) - 198 = 4 \cdot 252 - 5 \cdot 198.$$

Exercise 8.2

We have $d = \gcd(18, 42) = 6$, and $6 \mid 30$, so solutions exist and there are exactly six classes modulo 42. Divide by 6:

$$3x \equiv 5 \pmod{7}.$$

Since $3^{-1} \equiv 5 \pmod{7}$, $x \equiv 25 \equiv 4 \pmod{7}$. Therefore modulo 42 the solutions are

$$x \equiv 4, 11, 18, 25, 32, 39 \pmod{42}.$$

Exercise 8.3

The moduli 5, 7, 9 are pairwise coprime. First combine the first two conditions. Numbers congruent to 2 modulo 5 are 2, 7, 12, 17, 22, 27, 32, $\dots$; among these, $17 \equiv 3 \pmod{7}$. Thus $x \equiv 17 \pmod{35}$. Write $x = 17 + 35k$. Modulo 9,

$$17 + 35k \equiv 8 + 8k \equiv 4,$$

so $8k \equiv 5 \pmod{9}$. Since $8^{-1} \equiv 8$, $k \equiv 40 \equiv 4 \pmod{9}$. Hence

$$x \equiv 17 + 35 \cdot 4 = 157 \pmod{315}.$$

Exercise 8.4

For $1 \leq k \leq p-1$,

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

The denominator contains no factor p , while the numerator contains one, so $p \mid \binom{p}{k}$. Therefore the binomial theorem gives

$$(a+b)^p = a^p + b^p + \sum_{k=1}^{p-1} \binom{p}{k} a^k b^{p-k} \equiv a^p + b^p \pmod{p}.$$

Exercise 8.5

Assume only finitely many primes congruent to 3 (mod 4), say $p_1, \dots, p_r$. Let

$$N = 4p_1 \cdots p_r - 1.$$

Then $N \equiv 3 \pmod{4}$ and none of the p_i divides N . In the prime factorization of N , not all prime factors can be 1 (mod 4), because a product of numbers each congruent to 1 (mod 4) is 1 (mod 4). Also the prime 2 does not divide N . Hence some prime divisor q satisfies $q \equiv 3 \pmod{4}$. This q is not among the p_i , contradiction.

Exercise 8.6

The nonzero squares modulo 11 are obtained from $1^2, \dots, 5^2$ because $(11-k)^2 \equiv k^2$:

$$1, 4, 9, 5, 3.$$

Thus the residues are $\{1, 3, 4, 5, 9\}$. For the residue 3,

$$3^5 = 243 \equiv 1 \pmod{11}.$$

For the nonresidue 2,

$$2^5 = 32 \equiv -1 \pmod{11}.$$

This matches Euler's criterion.

A.9 Solutions to Chapter 9: probability

Exercise 9.1

Let the boxes be B_1, B_2, B_3 with priors $1/2, 1/3, 1/6$ and red probabilities $1/4, 1/2, 3/4$. The total probability of red is

$$\frac{1}{2} \frac{1}{4} + \frac{1}{3} \frac{1}{2} + \frac{1}{6} \frac{3}{4} = \frac{1}{8} + \frac{1}{6} + \frac{1}{8} = \frac{5}{12}.$$

Bayes gives

$$\mathbb{P}(B_1 | R) = \frac{1/8}{5/12} = \frac{3}{10}, \quad \mathbb{P}(B_2 | R) = \frac{1/6}{5/12} = \frac{2}{5}, \quad \mathbb{P}(B_3 | R) = \frac{1/8}{5/12} = \frac{3}{10}.$$

Exercise 9.2

Write $X = I_1 + \cdots + I_n$, where I_j indicates success on trial j . Then $\mathbb{E}I_j = p$, so by linearity

$$\mathbb{E}X = np.$$

Each I_j is Bernoulli, so $\text{Var}(I_j) = p(1 - p)$. Independence gives zero covariances, hence

$$\text{Var}(X) = \sum_j \text{Var}(I_j) = np(1 - p).$$

Exercise 9.3

For discrete independent variables,

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y).$$

Therefore

$$\mathbb{E}[XY] = \sum_{x,y} xy\mathbb{P}(X = x, Y = y) = \left(\sum_x x\mathbb{P}(X = x) \right) \left(\sum_y y\mathbb{P}(Y = y) \right) = \mathbb{E}X\mathbb{E}Y,$$

assuming absolute convergence guaranteed by the finite second moments.

Exercise 9.4

Chebyshev gives

$$\mathbb{P}(|X - 10| \geq 5) \leq \frac{4}{25} = 0.16.$$

No stronger universal bound follows from Chebyshev using only the stated mean and variance.

Exercise 9.5

For X uniform on $\{-1, 0, 1\}$, symmetry gives $\mathbb{E}X = 0$ and $Y = X^2$ takes values 0, 1. Also $XY = X^3$, whose expectation is 0. Hence

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y = 0.$$

But Y is determined by X , so they cannot be independent. For instance,

$$\mathbb{P}(X = 0, Y = 0) = \frac{1}{3},$$

while

$$\mathbb{P}(X = 0)\mathbb{P}(Y = 0) = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}.$$

Exercise 9.6

Let $\pi = (a, 1 - a)$. Solving $\pi P = \pi$ gives

$$0.8a + 0.3(1 - a) = a,$$

so $0.5a = 0.3$ and $a = 0.6$. Thus $\pi = (0.6, 0.4)$. Every row-stochastic matrix satisfies $P\mathbf{1} = \mathbf{1}$, so 1 is a right eigenvalue. Equivalently, 1 is an eigenvalue of P^T , which is why a stationary row vector is a left eigenvector with eigenvalue 1.

A.10 Solutions to Chapter 10: measure theory**Exercise 10.1**

Since

$$\mathbf{1}_{\liminf A_n} = \liminf_n \mathbf{1}_{A_n}$$

pointwise, Fatou's lemma gives

$$\mu(\liminf A_n) = \int \mathbf{1}_{\liminf A_n} d\mu \leq \liminf_n \int \mathbf{1}_{A_n} d\mu = \liminf_n \mu(A_n).$$

Thus the set inequality is exactly Fatou applied to indicator functions.

Exercise 10.2

If $\mu(X) < \infty$ and $\|f_n - f\|_\infty \rightarrow 0$, then

$$\|f_n - f\|_1 = \int |f_n - f| d\mu \leq \mu(X) \|f_n - f\|_\infty \rightarrow 0.$$

The boundedness assumption is not actually needed once uniform convergence to a finite-valued function is known and $\mu(X) < \infty$; the displayed estimate is sufficient whenever the sup norm is finite.

Exercise 10.3

For fixed $x > 0$, eventually $x > 1/n$, so $f_n(x) = 0$. Thus $f_n \rightarrow 0$ pointwise on $(0, 1)$. The integral is

$$\int_0^{1/n} n^2 x(1 - nx) dx.$$

Substitute $u = nx$, $dx = du/n$:

$$\int_0^1 u(1 - u) du = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Hence the integrals do not tend to 0. This is another moving-concentration example.

Exercise 10.4

Let $p, q > 1$ be conjugate. Convexity of $\phi(t) = t^p/p$ gives the supporting-line inequality at $t = 1$ after suitable scaling, but a direct optimization is clearer. Fix $a \geq 0$ and maximize $ab - b^q/q$ over $b \geq 0$. Its derivative is $a - b^{q-1}$, so the maximum occurs at $b = a^{p-1}$ because $(p-1)(q-1) = 1$. The maximum value is

$$a^p - \frac{a^{(p-1)q}}{q} = a^p \left(1 - \frac{1}{q}\right) = \frac{a^p}{p}.$$

Thus $ab - b^q/q \leq a^p/p$, which rearranges to Young's inequality.

Exercise 10.5

By the triangle inequality for the integral,

$$\left| \int f_n - \int f \right| \leq \int |f_n - f| = \|f_n - f\|_1 \rightarrow 0.$$

The converse fails. On $[0, 2\pi]$, take $f_n(x) = \sin(nx)$ and $f = 0$. Then $\int f_n = 0 = \int f$ for every n , but

$$\|f_n\|_1 = \int_0^{2\pi} |\sin(nx)| dx = 4$$

for every n , so there is no L^1 convergence.

Exercise 10.6

For every $\varepsilon > 0$, Markov's inequality applied to $|X_n - X|^2$ gives

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|X_n - X|^2 > \varepsilon^2) \leq \frac{\mathbb{E}|X_n - X|^2}{\varepsilon^2} \rightarrow 0.$$

Thus $X_n \rightarrow X$ in probability.

A.11 Solutions to Chapter 11: differential equations**Exercise 11.1**

The Picard operator is

$$(Tx)(t) = \int_0^t (1 + x(s)) ds.$$

Starting with $x_0 = 0$,

$$x_1 = t, \quad x_2 = t + \frac{t^2}{2}, \quad x_3 = t + \frac{t^2}{2} + \frac{t^3}{6},$$

and

$$x_4 = t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24}.$$

The limit is $e^t - 1$, the solution of $x' = 1 + x$, $x(0) = 0$.

Exercise 11.2

Write

$$A = 3I + N, \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

Then

$$e^{tA} = e^{3t}(I + tN) = e^{3t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Hence for $x(0) = (c_1, c_2)^T$,

$$x(t) = e^{3t} \begin{pmatrix} c_1 + tc_2 \\ c_2 \end{pmatrix}.$$

The factor t comes from the nilpotent shear in the Jordan block.

Exercise 11.3

For $x' = \lambda x$, $y' = \mu y$, solutions are $(c_1 e^{\lambda t}, c_2 e^{\mu t})$. If both eigenvalues are negative, the origin is an attracting node. If both are positive, it is a repelling node. If they have opposite signs, it is a saddle. If one eigenvalue is zero, linearization has a neutral direction and the equilibrium is nonhyperbolic; linear information alone does not classify perturbations in that direction.

Exercise 11.4

For $f(x) = rx(1 - x/K)$, equilibria satisfy $f(x) = 0$, so $x = 0$ and $x = K$. Since

$$f'(x) = r \left(1 - \frac{2x}{K} \right),$$

we have $f'(0) = r > 0$, so 0 is unstable, and $f'(K) = -r < 0$, so K is locally asymptotically stable.

Exercise 11.5

Along a solution,

$$\frac{d}{dt}H(x, v) = vv' + U'(x)x' = v(-U'(x)) + U'(x)v = 0.$$

Thus trajectories stay on level sets

$$\frac{1}{2}v^2 + U(x) = E.$$

The allowed region must satisfy $U(x) \leq E$, and the velocity is determined up to sign by $v = \pm \sqrt{2(E - U(x))}$. Energy therefore constrains turning points and qualitative motion without an explicit solution formula.

Exercise 11.6

Let x, y solve the same ODE with the same initial value and suppose f is Lipschitz in the state with constant L . Then

$$\|x(t) - y(t)\| \leq \int_{t_0}^t \|f(s, x(s)) - f(s, y(s))\| \, ds \leq L \int_{t_0}^t \|x(s) - y(s)\| \, ds.$$

Apply Gronwall with $A = 0$ to obtain $\|x(t) - y(t)\| = 0$. Thus the solutions coincide.

Compact recall deck and study plan

This appendix is deliberately terse. Use it only after working through the chapters. Cover the right-hand column and reconstruct each item aloud or on paper.

B.1 Linear algebra deck

Prompt	Reconstruction target
Vector space	A set with addition and scalar multiplication satisfying the linear axioms. Mental model: a world where effects can be superposed and rescaled.
Span	All finite linear combinations; the smallest subspace containing the generators.
Independence	No nontrivial linear combination vanishes. It is the absence of redundancy.
Basis	Independent plus spanning; exactly enough building blocks, hence unique coordinates.
Rank-nullity	$\dim V = \dim \ker T + \dim \operatorname{im} T$. Proof: extend a kernel basis to a basis of V and map the added vectors.
Quotient V/W	Declare differences lying in W invisible. Dimension drops by $\dim W$.
First isomorphism theorem	$V/\ker T \cong \operatorname{im} T$: a linear map forgets exactly its kernel.
Dual space	V^* is the space of linear measurements $V \rightarrow F$.
Determinant	Alternating multilinear oriented volume-scaling factor. Multiplicative under composition.
Eigenvector	A direction that the operator does not turn; only scales.
Diagonalizable	There is a basis of eigenvectors. Dynamics decouple into scalar modes.
Orthogonal projection	Closest point in a subspace; residual is perpendicular to the subspace.
Self-adjoint	$T = T^*$. Spectral theorem gives an orthonormal eigenbasis.

Jordan block	Scaling λI plus nilpotent shear. Polynomial factors record failure of diagonalization.
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B.2 Multivariable calculus deck

Prompt	Reconstruction target
Differentiability	$F(a + h) = F(a) + DF(a)h + o(\ h\)$. Derivative is the best first-order linear model.
Jacobian	Matrix of $DF(a)$ in standard coordinates, not the definition of differentiability.
Chain rule	$D(G \circ F) = DG \circ DF$. Local linear models compose.
Gradient	Inner-product representative of the differential: $Df(h) = \langle \nabla f, h \rangle$.
Hessian	Matrix of the second derivative; at a critical point it gives the first quadratic shape.
Lagrange multipliers	At a constrained extremum, the differential vanishes on tangent directions, so gradient lies in the normal span.
Inverse function theorem	Invertible derivative $\Rightarrow$ locally invertible nonlinear map. Proof mechanism: normalize derivative and use contraction.
Implicit function theorem	Nondegeneracy in selected variables gives local permission to solve for those variables as functions of the others.

B.3 Topology deck

Prompt	Reconstruction target
Topology	Collection of open sets closed under arbitrary unions and finite intersections and containing $\emptyset, X$.
Basis	Elementary open neighborhoods whose unions generate all opens.
Closure	Points every neighborhood of which meets the set.
Continuity	Inverse images of open sets are open. In metric spaces this is equivalent to ε - δ .
Homeomorphism	Continuous bijection with continuous inverse; topological sameness.
Compactness	Every open cover has a finite subcover. Deep role: convert local control into finite/global control.
Connectedness	No separation into two disjoint nonempty open sets. Continuous images preserve it.
Product topology	Basic opens restrict finitely many coordinates; observations have finite coordinate resolution.
Quotient topology	U is open downstairs iff its full preimage is open upstairs. Designed for gluing.
Hausdorff	Distinct points have disjoint neighborhoods. Consequences: unique limits; compact subsets closed.

Homotopy	Continuous deformation of maps. Coarser equivalence than homeomorphism.
Fundamental group	Homotopy classes of based loops under concatenation; algebraic probe of one-dimensional holes.

B.4 Discrete mathematics and graph theory deck

Prompt	Reconstruction target
Addition principle	Count disjoint alternatives by summing.
Multiplication principle	Count tuples of successive choices by multiplying.
Bijection	Explain equality of cardinalities by explicit reversible pairing.
Pigeonhole	A map from more objects to fewer signatures cannot be injective.
Inclusion-exclusion	Correct overcount by alternating intersections. Proof: inspect one element belonging to r sets.
Recurrence	Express size- n objects through smaller objects by a structural first or last decision.
Invariant	Quantity unchanged by legal moves; differing invariant values obstruct reachability.
Monovariant	Quantity moving strictly one way; proves termination or impossibility of cycling.
Equivalence relation	Reflexive, symmetric, transitive rule of indistinguishability; same data as a partition.
Partial order	Reflexive, antisymmetric, transitive relation; not every pair need be comparable.
Generating function	Encode a_n as coefficient of x^n ; shifts become multiplication by x , convolution becomes product.
Handshake lemma	$\sum_v \deg v = 2 E $. Double count incidences.
Tree	Connected acyclic graph; equivalent to unique paths and minimal connectivity with $n - 1$ edges.
Euler trail	Uses every edge once. Criterion: 0 or 2 odd-degree vertices in a connected graph.
Bipartite	Two-colorable; equivalent to no odd cycle.
Hall theorem	Matching saturating L iff every $S \subseteq L$ has $ N(S) \geq S $.
Planarity	Euler formula $V - E + F = 2$ for connected planar embeddings; edge bounds create obstructions.

B.5 Number theory deck

Prompt	Reconstruction target
Division algorithm	$a = bq + r$, $0 \leq r < b$. Existence from well-ordering; uniqueness from remainder bounds.
Euclidean algorithm	Preserve gcd while strictly decreasing the remainder.

Bezout identity	$ax + by = \gcd(a, b)$. The gcd generates the ideal (a, b) .
Euclid lemma	Prime dividing a product divides a factor. Proof uses Bezout.
Unique factorization	Existence by strong induction; uniqueness by repeated Euclid lemma.
Congruence	$a \equiv b \pmod{n}$ iff $n \mid a - b$; equality in $\mathbb{Z}/n\mathbb{Z}$.
Linear congruence	$ax \equiv b \pmod{n}$ solvable iff $\gcd(a, n) \mid b$.
CRT	Coprime modular coordinates are independent; $\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$.
Euler theorem	$a^{\varphi(n)} \equiv 1 \pmod{n}$ for units; Lagrange theorem in $(\mathbb{Z}/n\mathbb{Z})^\times$.
Quadratic residue	Nonzero square modulo p ; Euler criterion tests via exponent $(p - 1)/2$.

B.6 Probability and measure deck

Prompt	Reconstruction target
Probability space	$(\Omega, \mathcal{F}, \mathbb{P})$: outcomes, measurable events, countably additive normalized measure.
Conditional probability	Restrict to B and renormalize: $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$.
Independence	$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$; learning one gives no information about the other.
Random variable	Measurable numerical probe of a random outcome.
Expectation	Linear average / integral. Independence is unnecessary for linearity.
Variance	Mean square distance from the mean. For sums, covariance is the correction term.
Markov inequality	Nonnegative mass above a is bounded by mean divided by a .
Chebyshev	Apply Markov to squared centered deviation.
Weak law	Variance of iid average is σ^2/n ; Chebyshev forces concentration.
Sigma-algebra	Sets stable under complements and countable unions; the natural event language for sequential analysis.
Measure	Countably additive mass.
Continuity from below	Increasing sets: measure of union is limit of measures.
Continuity from above	Decreasing sets with finite first measure: measure of intersection is the limit.
Measurable function	Threshold events $\{f > a\}$ are measurable.
Lebesgue integral	Build from indicators to simple functions to nonnegative measurable functions by supremum.
MCT	Increasing nonnegative functions: integral commutes with limit.
Fatou	Integral of $\liminf \leq \liminf$ of integrals.
DCT	Pointwise convergence plus one integrable dominator allows limit through the integral.
L^p	Integrable p th power, modulo almost-everywhere equality; norm measures average magnitude.

B.7 ODE and dynamics deck

Prompt	Reconstruction target
Vector field	Assigns instantaneous velocity to each state. Solutions follow the arrows tangentially.
Integral equation	$x(t) = x_0 + \int f(s, x(s))ds$. Converts ODE to fixed-point problem.
Contraction theorem	Complete space + strict contraction $\Rightarrow$ unique fixed point, found by iteration.
Picard-Lindelof	Continuity + local Lipschitz in state $\Rightarrow$ local existence and uniqueness.
Gronwall	Integral inequality $u \leq A + B \int u$ gives $u \leq Ae^{Bt}$. Controls error growth.
Matrix exponential	e^{tA} is the flow of $x' = Ax$. Eigenstructure determines modes.
Linearization	Near equilibrium a , $h' = Df(a)h + o(\ h\)$.
Conserved quantity	$\nabla H \cdot f = 0$ along flow; level sets constrain trajectories.
Lyapunov function	Scalar quantity decreasing along trajectories; replaces explicit solution in stability analysis.

B.8 Proof-mechanism deck

When a proof stalls, classify the desired conclusion before searching for a theorem.

Desired conclusion	High-yield mechanism
Dimension identity	Build a basis adapted to kernel/intersection; use rank-nullity.
Best approximation	Produce an orthogonal decomposition and use Pythagoras.
Local nonlinear theorem	Subtract the linear part; normalize; control the remainder.
Uniform conclusion from pointwise local data	Look for compactness and a finite subcover.
Connected-image conclusion	Pull a hypothetical separation back through continuity.
Counting identity	Partition the same family differently or build a bijection.
Impossible configuration	Search for parity, coloring, modular, or monotone invariant.
Global assignment	Test Hall-type capacity inequalities; induct on a tight subset.
Congruence inverse	Run extended Euclid / Bezout.
Tail probability	Turn the tail event into an indicator inequality; use Markov/Chebyshev.
Interchange limit and integral	Identify monotonicity or domination; do not use pointwise convergence alone.
ODE uniqueness	Subtract two solutions, use Lipschitz, then Gronwall or contraction.

B.9 A twelve-day reconstruction schedule

This is a default, not a deadline.

Day	Focus	Required output
1	Diagnostic	Closed-book attempt; classify each miss as vocabulary, theorem choice, or execution.
2	Linear algebra I	Reconstruct rank-nullity and the quotient/isomorphism picture; do all exercises.
3	Linear algebra II	Projection, spectral theorem architecture, one Jordan computation.
4	Multivariable calculus	Derivative definition, chain rule proof, IFT/implicit theorem architecture.
5	Topology I	Open sets, continuity, closure, compactness; reconstruct uniform-continuity proof.
6	Topology II	Connectedness, products, quotients, Hausdorff, homotopy.
7	Discrete math	Bijections, inclusion-exclusion, recurrences, invariants, generating functions.
8	Graph theory	Trees, Euler trails, bipartiteness, Hall, planarity.
9	Number theory	Euclid/Bezout, congruences, CRT, Euler theorem.
10	Probability	Conditioning, expectation, variance, inequalities, weak law.
11	Measure theory	Sigma-algebras, MCT/Fatou/DCT; explain one counterexample to naive limit interchange.
12	ODEs + synthesis	Picard-Lindelof mechanism, linear systems, stability, then the final mastery questions.

B.10 Final readiness rubric

A topic is ready for further study when you can do all four of the following without recognition cues:

1. state the central definitions accurately and say what problem each solves;
2. generate a concrete example, a nonexample, and a near-miss;
3. reconstruct at least one theorem proof from a one-sentence proof architecture;
4. compute a standard example and explain what the computation means structurally.

Deep intuition

The goal of a reboot is not to freeze mathematics in memory. It is to recover enough internal structure that forgotten details can be regenerated. A theorem you truly understand is not stored as a paragraph; it is stored as a network of reasons.

Hypothesis audit: statements that are almost true

A mature refresher should train not only theorem recall but theorem boundaries. The statements below are deliberately close to correct. Each repair identifies a hypothesis that carries real mathematical work.

C.1 Linear algebra

Common pitfall

False: A repeated eigenvalue makes an operator non-diagonalizable.

Repair: Repeated eigenvalues are compatible with diagonalizability; $2I$ is the simplest example. Failure occurs when the eigenspaces do not supply enough independent eigenvectors. Geometric multiplicity, not repetition by itself, is decisive.

Common pitfall

False: If $A^T A$ is invertible, then every system $Ax = b$ has an exact solution.

Repair: $A^T A$ invertible means the columns of A are independent. It guarantees a unique least-squares solution, not that b lies in the column space.

Common pitfall

False: Every vector space is naturally isomorphic to its dual.

Repair: Finite-dimensional V and V^* have equal dimension, so an isomorphism exists after choices. A canonical map exists instead from V to V^{**} . An inner product supplies a natural identification with the dual relative to that additional structure.

C.2 Multivariable calculus

Common pitfall

False: If all partial derivatives exist at a point, the function is differentiable there.

Repair: Partial derivatives test only coordinate directions. Differentiability requires one linear approximation controlling all directions simultaneously. Continuity of the first partial derivatives in a neighborhood is a standard sufficient condition.

Common pitfall

False: A zero Hessian at a critical point means the point is neither a maximum nor a minimum.

Repair: The second derivative test is inconclusive when the Hessian is semidefinite or zero. Higher-order terms may decide the geometry, as with x^4 .

Common pitfall

False: A nonzero Jacobian determinant makes a map globally invertible.

Repair: The inverse function theorem is local. The exponential-polar map $(x, y) \mapsto (e^x \cos y, e^x \sin y)$ has positive Jacobian determinant everywhere but is periodic in y .

C.3 Topology

Common pitfall

False: Compact means closed and bounded.

Repair: In $\mathbb{R}^n$ with its usual topology, Heine-Borel says compact is equivalent to closed and bounded. In a general metric or topological space this characterization can fail; open-cover compactness is the intrinsic definition.

Common pitfall

False: A continuous bijection is a homeomorphism.

Repair: The inverse must also be continuous. A useful sufficient condition is: continuous bijection from compact to Hausdorff is automatically a homeomorphism.

Common pitfall

False: Sequential compactness and compactness are always equivalent.

Repair: They coincide in metric spaces. General topological spaces may require nets or filters to capture compactness.

Common pitfall

False: The quotient topology is obtained by declaring images of open sets open.

Repair: Images of open sets under a quotient map need not be open. A set U downstairs is open exactly when its full preimage is open upstairs.

C.4 Discrete mathematics and graph theory

Common pitfall

False: Pairwise disjoint cases are needed for the multiplication principle.

Repair: Disjoint alternatives support addition. Multiplication applies when an object is encoded uniquely by successive choices. Mixing these two decompositions is a common counting error.

Common pitfall

False: A graph with $n - 1$ edges is a tree.

Repair: One also needs connectedness or acyclicity. A disconnected graph can have $n - 1$ edges while containing a cycle in one component and isolated structure elsewhere.

Common pitfall

False: Every graph with all even degrees has an Euler circuit.

Repair: The non-isolated part must be connected. Parity is necessary but cannot bridge disconnected components.

Common pitfall

False: The greedy coloring algorithm always uses $\chi(G)$ colors.

Repair: Greedy coloring gives an upper bound, at most $\Delta + 1$, and can be far from optimal under a poor vertex order.

C.5 Number theory

Common pitfall

False: Every nonzero residue modulo n has a multiplicative inverse.

Repair: $[a]$ is a unit in $\mathbb{Z}/n\mathbb{Z}$ exactly when $\gcd(a, n) = 1$. All nonzero residues are units precisely when n is prime.

Common pitfall

False: $ax \equiv b \pmod{n}$ has a unique solution whenever it is solvable.

Repair: If $d = \gcd(a, n)$ divides b , there are exactly d solution classes modulo n . Uniqueness occurs when $d = 1$.

Common pitfall

False: Euler's theorem says $a^{\varphi(n)} \equiv 1 \pmod{n}$ for every integer a .

Repair: The theorem requires $\gcd(a, n) = 1$, because it is a statement inside the unit group $(\mathbb{Z}/n\mathbb{Z})^\times$.

C.6 Probability and measure

Common pitfall

False: Disjoint events are independent.

Repair: Nontrivial disjoint events are dependent: occurrence of one makes the other impossible. Independence means intersection probability factors.

Common pitfall

False: Zero covariance implies independence.

Repair: Independence implies zero covariance under finite second moments, but the converse fails except in special families such as jointly Gaussian variables.

Common pitfall

False: If $f_n \rightarrow f$ pointwise, then $\int f_n \rightarrow \int f$.

Repair: Pointwise convergence does not prevent mass from concentrating or escaping. Monotone convergence, dominated convergence, or another uniform-integrability condition is needed.

Common pitfall

False: For decreasing measurable sets E_n , one always has $\mu(\cap E_n) = \lim \mu(E_n)$.

Repair: Continuity from above requires $\mu(E_1) < \infty$. Without it, both sides can behave differently, as $[n, \infty) \downarrow \emptyset$ under Lebesgue measure shows.

Common pitfall

False: Almost-everywhere equality means two functions are literally the same function.

Repair: They may differ on a null set. L^p spaces deliberately quotient out this distinction, so an L^p “function” is formally an equivalence class.

C.7 Differential equations

Common pitfall

False: Continuity of the vector field guarantees a unique solution through each initial point.

Repair: Continuity supports existence under Peano-type results, but uniqueness needs more control, such as a local Lipschitz condition in the state variable.

Common pitfall

False: Eigenvalues of the linearization always determine nonlinear stability.

Repair: If no eigenvalue has zero real part, linearization is robust. When zero real parts occur, higher-order nonlinear terms can decide stability and the linearized system may be inconclusive.

Common pitfall

False: Separating variables is harmless algebra.

Repair: Dividing by a state-dependent factor can discard equilibrium solutions where that factor vanishes. Isolate singular cases before division.

C.8 A final correctness habit

Before applying a theorem, perform a three-line hypothesis audit:

1. Write the theorem in a form that includes every hypothesis.
2. Match each hypothesis to a fact in the current problem.
3. Ask what counterexample appears if the least obvious hypothesis is removed.

Deep intuition

Hypotheses are not bureaucratic decorations around a theorem. They are the walls that keep known counterexamples out. Understanding a theorem deeply includes knowing which failure mode each wall prevents.